Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Tomato

1.1 Late Blight

Disease Name and Causal Organism:

- **Disease Name:** Late Blight.
- **Causal Organism:** Phytophthora infestans (also referenced in documentation as Phytopthora infestans). Four races of the pathogen have been reported.

Symptoms and Signs:

- **Leaf Symptoms:** Initial symptoms appear as bending down of the petioles of infected leaves and small, irregular, greenish-black or dark brown water-soaked spots. Spots enlarge rapidly into large, dark brown, paper-like patches. Under humid or wet weather conditions, a white sporulating fungal growth or white mold/fuzz develops near the margins on the lower surface of the leaves.
- **Stem Symptoms:** Dark, greasy-looking patches or brown streaks develop on stems and petioles, which spread rapidly, killing the plant tissue or causing rapid blighting of the entire foliage.
- **Fruit Symptoms:** Symptoms on fruits appear as large, firm, irregular, dark brownish-green or greasy blotches, as well as sunken brown spots. These lesions occur mostly on the upper half or shoulders of the fruit where airborne spores land. Under humid conditions, white moldy fungal growth also develops on fruit lesions.
- **General Signs:** Severe infections lead to rapid foliage destruction, stem girdling, and complete crop collapse, capable of destroying entire fields within a few days.

Disease Development and Spread:

- **Primary Inoculum and Survival:** The fungal pathogen survives on volunteer and home garden tomato and potato plants, in potato cull piles, in infected plant debris, and on solanaceous weeds.
- **Dispersal Mechanisms:** Airborne spores spread rapidly over long distances dispersed by wind and high-wind storms, as well as secondary transmission through rain splash.
- **Infection Cycle:** Spores land on plants and infect leaves, stems, and fruits. Under favorable conditions, the infection spreads widely across fields within 4 to 5 days, causing severe outbreaks.

Conditions Favourable for Disease Development:

- **Temperature:** Cool weather conditions, with ideal temperatures ranging between 15°C and 18°C.
- **Humidity and Moisture:** High relative humidity exceeding 90%, persistent dew, and continuous wet or rainy weather.
- **Wind:** High wind speeds that disperse airborne spores across fields.

Diagnosis and Identification:

- **Field Visual Diagnosis:** Regular crop inspection to detect early visual warning signs, including petiole drooping, water-soaked leaf lesions with white undersurface sporulation during humid conditions, dark greasy stem streaks, and firm brown shoulder blotches on fruit.
- **Blight Forecasting Systems:** Monitoring local weather forecasts and utilizing blight forecasting systems to predict disease outbreaks and timing.
- **Laboratory Identification:** Not specified in the provided sources.

Prevention and Cultural Practices:

- **Resistant Varieties:** Cultivate resistant varieties or hybrids such as Arka Abhed F1.
- **Crop Rotation:** Avoid planting tomatoes in the same field consecutively every year, and avoid planting tomatoes near or following potato crops.
- **Clean Planting Material:** Obtain seeds from disease-free areas and utilize disease-free seeds and transplants.
- **Field Sanitation:** Practice field hygiene by collecting, removing, and destroying infected plant debris, shoots, and solanaceous weed hosts to reduce pathogen inoculum.
- **Plant Spacing and Staking:** Ensure adequate spacing between plants and practice staking to promote proper airflow and decrease canopy humidity buildup.
- **Soil Mulching:** Use black polythene mulch (30-micron Low Density Polyethylene) buried at both ends to a depth of 10 cm to reduce soil moisture, prevent disease spread, and suppress weeds.
- **Water and Cultural Management:** Improve field drainage, provide light irrigation, avoid overhead irrigation, plant windbreaks, and avoid late sowing during the monsoon season.

Biological and Chemical Control:

- **Biological Control:** Not specified in the provided sources for Late Blight specifically.
- **Seed Treatment:** Treat seeds prior to planting with Thiram at a rate of 2 to 3 g per kg of seed.

- **Organic and Copper-Based Fungicides:** Apply Copper Oxychloride 50% WP at a dosage of 2.5 g per liter of water (or 1000 g in 300–400 liters of water per acre), delivered as 6 preventive sprays at 6-day intervals combined with soil mulching. Alternatively, apply Copper sulphate 2.62% SC at 400 ml in 200 liters of water per acre.
- **Preventive Foliar Chemical Sprays:** Apply Captafol at 2 g per liter of water or Mancozeb (Dithane M-45) at 2 g per liter of water (or Mancozeb 75% WP at 600–800 g in 300 liters of water per acre, or Mancozeb 35% SC at 200 g in 200 liters of water per acre) at 15-day intervals, beginning 30 days after transplanting.
- **Additional Recommended Chemical Fungicides:** Documented chemical options include Captan 50% WP at 1000 g in 300–400 liters of water per acre (or Captan 75% WP at 666.8 g in 400 liters of water per acre); Famoxadone 16.6% + Cymoxanil 22.1% SC at 200 g in 200 liters of water per acre; Cymoxanil 8% + Mancozeb 64% WP at 600 g in 200–300 liters of water per acre; Azoxystrobin 23% SC at 200 ml in 200 liters of water per acre; Cyazofamid 34.5% SC at 80 ml in 200 liters of water per acre; Metiram 75% WG at 1000 g in 200–300 liters of water per acre (or Metiram 55% + Pyraclostrobin 5% WG at 600–700 g in 200 liters of water per acre); Tebuconazole 50% + Trifloxystrobin 25% WG at 0.75 g per liter; and Zineb 75% WP at 600–800 g in 300–400 liters of water per acre.

Integrated Disease Management:

- **Holistic Management Program:** Combine resistant varieties (Arka Abhed F1), clean seed selection, crop rotation with non-host crops, wide plant spacing, staking, field sanitation, black polythene mulching, improved drainage, and a scheduled spray program of copper oxychloride or labeled fungicides linked to blight forecasting.

Monitoring and Early Detection:

- **Regular Crop Monitoring:** Routinely check plants for initial water-soaked leaf lesions, petiole drooping, stem discoloration, and fruit shoulder spots, particularly during cool, wet weather.
- **Early Preventive Intervention:** Initiate preventive organic or chemical fungicide applications prior to or at the earliest sign of disease appearance.

Effects on Crop Growth, Fruit Quality, and Yield:

- **Effects on Plant Growth:** Causes rapid blighting of foliage, stem girdling, and severe tissue necrosis, resulting in complete plant mortality.
- **Effects on Fruit Quality:** Produces firm, large, dark brown greasy lesions and sunken moldy blotches that destroy the visual appearance, structure, and market value of the fruits.
- **Effects on Yield:** Uncontrolled outbreaks cause widespread crop destruction within 4 to 5 days, resulting in massive economic losses. Adopting recommended integrated management practices reduces disease impact by up to 92.04% and increases crop yields up to 9.25 t/ha.

Conflicting and Uncertain Recommendations:

- **Spray Schedule and Dose Variations:** Recommendations across source documents vary regarding spray timing and frequency, ranging from a 6-spray preventive schedule of Copper Oxychloride at 6-day intervals paired with mulching, to 15-day spray intervals of Mancozeb or Captafol initiated 30 days post-transplanting.

1.2 Early Blight

- **Disease Name and Causal Organism:** Early Blight, caused by the fungal pathogens *Alternaria solani* and *Alternaria alternata* f. sp. *lycopersici*. The fungal mycelium is septate, branched, light brown, and darkens with age. Conidiophores are dark, and conidia are dark-colored, beaked, muriform, and borne singly.

- **Symptoms and Signs:** Symptoms can occur at any growth stage on the foliage, stems, and fruits. On foliage, small black lesions initially appear on older leaves, enlarging up to one-fourth inch or more with concentric rings forming a bull's-eye or target-board pattern in the center, surrounded by yellow chlorotic tissue. Severe leaf infection causes foliage to dry rapidly and leads to complete defoliation. Stem lesions are dark brown, elongated, sunken, and display concentric rings; when occurring near the soil line, they cause collar rot that girdles stems and kills plants or transplants. Fruit lesions generally develop near the calyx or stem attachment as large, dark, sunken, leathery areas with concentric rings that can cover nearly the entire fruit.

- **Disease Development and Spread:** The pathogen survives between seasons in infected plant debris in or on the soil for one to three years, as well as on seeds, volunteer tomato plants, and solanaceous weeds. Secondary spread occurs rapidly through spores transported by wind, rain splashes, water, insects, animals, machinery, and field workers.

- **Conditions Favourable for Disease Development:** Disease development is favoured by warm, rainy, and wet weather, with temperatures ranging between 24–29 °C (75–84 °F), high humidity, persistent dew, or sprinkler irrigation. Poor plant vigour resulting from nutrient inadequacy or late-season growth also increases susceptibility.

- **Diagnosis and Identification:** Diagnosed by the characteristic bull's-eye pattern of concentric rings within dark lesions on leaves, stems, and fruits, alongside chlorotic yellowing surrounding leaf spots starting on lower foliage.

- **Prevention and Cultural Practices:** Cultural preventive measures include growing resistant or tolerant cultivars, removing and destroying infected crop debris, changing nursery bed locations every season, and deep summer ploughing to expose soil-borne inoculum. Growers should practice crop rotation with non-solanaceous crops such as cereals, legumes, cucurbits, or cruciferous vegetables, avoid planting overlapping tomato crops in adjacent fields, maintain field sanitation, eradicate solanaceous weeds, ensure proper fertilization, and manage irrigation to avoid prolonged leaf wetness.

- **Biological and Chemical Control:** Biological control includes seed treatment with *Trichoderma viride* 1% WP at 9 g/kg of seed (or *Pseudomonas fluorescens* and *Trichoderma harzianum* formulations at 15–20 g/kg or 200 g/kg of seed), dipping seedling roots in bio-control agent solution at 5 g/L for 10–15 minutes, and applying bio-agents enriched in farmyard manure or vermicompost to the soil. Chemical control involves dipping seedling roots in fungicide solutions before transplanting and applying foliar sprays of protective or systemic fungicides.

- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Documented chemical treatments include foliar sprays of Azoxystrobin 23% SC at 200 ml in 200 L water/acre; Captan 50% WP at 1000 g in 300–400 L water/acre or Captan 75% WP at 666.8 g in 400 L water/acre; Copper Oxychloride 50% WP at 1000 g in 300–400 L water/acre (or 2.5 g/L to 3 g/L); Copper Sulphate 2.62% SC at 400 ml in 200 L water/acre; Iprodione 50% WP at 600 g in 200 L water/acre; Kitazin 48% EC at 80 ml in 80 L water/acre; Mancozeb 35% SC at 200 g in 200 L water/acre or Mancozeb 75% WG/WP at 400 g to 800 g in 200–300 L water/acre (or 0.2% spray); Pyraclostrobin 20% WG at 150–200 g in 200 L water/acre; Zineb 75% WP at 600–800 g in 300–400 L water/acre; Ziram 80% WP at 600–800 g in 300–400 L water/acre; Famoxadone 16.6% + Cymoxanil 22.1% SC at 200 g in 200 L water/acre; Metiram 55% + Pyraclostrobin 5% WG at 600–700 g in 200 L water/acre; Metiram 75% WG at 1000 g in 200–300 L water/acre; Cymoxanil 8% + Mancozeb 64% WP at 600 g in 200–300 L water/acre; Tebuconazole 50% + Trifloxystrobin 25% WG at 0.75 g/L; Difolatan / Captafol at 0.2%; Dithane M-45 at 0.2%; and Bavistin at 0.1%.

- **Integrated Disease Management:** Integrates the use of resistant or tolerant cultivars, deep summer ploughing, crop rotation with non-host crops, nursery solarization, field sanitation, seed and seedling biological treatments, and need-based chemical sprays combined with blight forecasting systems.

- **Monitoring and Early Detection:** Regular field scouting of older foliage for early black lesions, combined with blight forecasting systems to determine effective fungicide spray timing.

- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes yield losses up to 78%. It severely impairs crop growth through extensive foliage loss and plant death from stem girdling (collar rot), while fruit lesions cause large, dark, leathery decay that significantly lowers fruit quality and commercial value.

- **Conflicting or Uncertain Recommendations:** Dosing and formulation recommendations for chemical controls vary across sources depending on product concentration and application guidelines (for example, Mancozeb rates range from 200 g/acre for 35% SC to 400–800 g/acre for 75% WP/WG or 0.2% sprays; Copper Oxychloride is cited at both 2.5 g/L and 1000 g per 300–400 L water per acre).

- **Other Relevant Disease-Specific Information:** Not specified in the provided sources.

1.3 Fusarium Wilt

- **Disease Name and Causal Organism:** Fusarium wilt is caused by the fungal pathogen *Fusarium oxysporum* f. sp. *lycopersici*. The fungus produces septate, hyaline mycelium, one-celled hyaline microconidia ranging from ovoid to ellipsoid in shape, macroconidia, and persistent chlamydospores.

- **Symptoms and Signs:** Initial symptoms present as clearing of the veinlets and chlorosis (yellowing) of the leaves. In seedlings and young plants, symptoms include stunting, yellowing of older leaves and cotyledons, petiole drooping, and plant death. In field crops, yellowing begins on lower leaves first, followed by wilting and leaflet death, frequently displaying a "yellow flag" appearance where yellowing occurs on only one side of a leaf or branch. Affected plants undergo diurnal wilting on sunny days, experience stunting, and eventually wilt completely and die within a few days while leaves often remain attached. Cutting the stem longitudinally or diagonally, or snapping off branches, reveals a characteristic red-brown to dark brown discoloration of the internal vascular tissue.

- **Disease Development and Spread:** The pathogen is both soil-borne and seed-borne. It survives in the soil for several years (over 10 years) as resting chlamydospores or as saprophytically growing mycelium in infected plant debris. Infection occurs through root wounds caused by cultivation, secondary root formation, or nematode feeding. Pathogen spread occurs via infected soil, diseased seedlings, wind-borne spores, surface drainage and irrigation water, farm machinery, and contaminated agricultural implements.

- **Conditions Favourable for Disease Development:** Warm soil temperatures around 28 °C (82 °F) favour rapid disease development. Disease severity is also enhanced by high soil levels of micronutrients, phosphorus, and ammoniacal nitrogen.

- **Diagnosis and Identification:** Field identification relies on observing visual markers such as veinlet clearing, petiole drooping, "yellow flag" foliage chlorosis, and diurnal wilting. Diagnostic confirmation is achieved by inspecting cut stems or snapped branches for red-brown vascular discoloration extending up the plant stem.

- **Prevention and Cultural Practices:** Inspect nurseries and field crops regularly, removing and destroying wilt-infected plants immediately to reduce inoculum build-up. Practice crop rotation with non-host crops, such as cereals, to reduce soil pathogen levels, maintain strict field sanitation, and utilize clean, pathogen-free seed beds.

- **Biological and Chemical Control:** Biological management involves seed treatment and soil application using beneficial fungal biocontrol agents like *Trichoderma viride*. Chemical control includes fungicide seed treatments, localized spot drenching, and soil drenching using systemic or contact fungicides.

- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Biological seed treatment using *Trichoderma viride* 1% WP @ 9 g/kg of seed, or chemical seed treatment with Thiram @ 2.5 kg/ha prior to planting. Root zone biological application by thoroughly mixing 2.5 kg of *Trichoderma viride* 1% WP

in 150 kg of compost or farmyard manure and applying to the field after sowing or transplanting. Localized spot drenching or nursery bed drenching with Carbendazim (0.1%) or Carbendazim 50% WP @ 1 g/litre prior to planting. Soil drenching around affected areas with Captan 50% WP @ 1 g/litre after removing infected plants.

- **Integrated Disease Management:** Combines host plant resistance, biological seed and root zone applications (*Trichoderma viride*), strict field sanitation, roguing of infected plants, crop rotation with non-host cereal crops, and targeted chemical seed treatments or soil drenching.
- **Monitoring and Early Detection:** Perform routine monitoring of nursery beds and young field plantings for early visual symptoms such as leaf veinlet clearing, petiole drooping, and chlorosis of lower foliage.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes severe plant stunting, total systemic wilting, and premature plant death, leading to substantial crop and yield loss. Specific effects on fruit quality are not specified in the provided sources.
- **Other Relevant Disease-Specific Information:** Utilizing resistant tomato varieties is recognized as the most effective control strategy. Conflicting source information exists regarding pathogen races: one source notes two distinct races differing in virulence, whereas another source specifies three races (races 1, 2, and 3).

1.4 Bacterial Wilt

- **Disease Name and Causal Organism:** Bacterial wilt (also known as Southern Bacterial Wilt) is caused by the bacterial pathogen *Ralstonia solanacearum* (synonyms in sources include *Pseudomonas solanacearum* and *Burkholderia solanacearum*). The bacterium is gram-negative, rod-shaped (often occurring in pairs), and motile with 1–4 flagella. Five races of the pathogen have been reported.
- **Symptoms and Signs:** Initial symptoms present as drooping of lower leaves without foliar yellowing, or lower leaves dropping prior to wilting. Fully grown plants undergo rapid, complete, and sudden wilting without prior signs of stress. Internally, the pathogen invades the vascular system, cortex, and pith, causing yellow-brown to dark brown discoloration and hollow tissue decay. Freshly cut stems exude a slimy, white to gray bacterial exudate.
- **Disease Development and Spread:** The pathogen is soil-borne with an extremely broad host range exceeding 200 plant species, including solanaceous crops like eggplant, chili, potato, tobacco, and groundnut. It survives in fallow soil for up to 3 years and persists long-term in cultivated soil, plant debris, and alternate host weeds. Infection occurs through root wounds created by transplanting, cultivation, secondary root emergence, or nematode feeding, as well as chewing insects. Spread occurs through contaminated irrigation water flowing between fields, infested soil on tools and equipment, and infected transplants.

- **Conditions Favourable for Disease Development:** Disease expression is favoured by high soil moisture, wet or rainy weather, high relative humidity or persistent dew, and warm soil and environmental temperatures around 28–35 °C (84–95 °F).
- **Diagnosis and Identification:** Field diagnosis relies on observing sudden and complete wilting of plants without leaf chlorosis. Diagnostic confirmation is made by cutting the stem to inspect for yellow-brown vascular discoloration and placing a freshly cut stem piece upright in clear water to observe a white, milky stream of bacterial ooze emerging from the cut ends.
- **Prevention and Cultural Practices:** Use pathogen-free seed beds and certified disease-free seedlings. Avoid mechanical damage to roots during transplanting. Restrict irrigation water runoff from infected fields to healthy fields. Maintain strict field sanitation, rogue out infected plants, and practice deep summer ploughing or soil solarization. Rotate crops with non-host plants, particularly paddy or specific rotations such as cowpea-maize-cabbage, okra-cowpea-maize, maize-cowpea-maize, or finger millet-eggplant.
- **Biological and Chemical Control:** Biological control includes soil incorporation of Neem cake. Chemical management involves soil fumigation, soil application of bleaching powder, grafting onto resistant rootstocks, seedling root dip treatments using bactericides (Streptocycline), and foliar sprays of copper-based fungicides.
- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Biological soil treatment by applying Neem cake @ 100 kg/acre (500 kg/ha) to the main field prior to transplanting. Chemical soil treatment by applying Bleaching powder @ 10 kg/ha to suppress soil bacterial populations, or soil fumigation. Seedling chemical protection by dipping roots for 30 minutes in a Streptocycline solution (streptomycin sulfate 9% + tetracycline hydrochloride 1% SP) @ 1 g per 40 litres of water before transplanting. Foliar chemical control by spraying Copper oxychloride 50% WP @ 2–3 g/litre of water.
- **Integrated Disease Management:** Combines host plant resistance using tolerant or resistant varieties, crop rotation with non-host crops like paddy or cereal-legume sequences, strict nursery and field sanitation, restricting contaminated irrigation runoff, soil application of Neem cake @ 100 kg/acre and Bleaching powder @ 10 kg/ha, grafting onto resistant rootstocks, and preventive seedling root dipping in Streptocycline solution (1 g/40 litres).
- **Monitoring and Early Detection:** Routinely inspect nursery beds and field crops for early visual signs such as lower leaf drooping. Confirm suspected infections early by performing the water-glass bacterial streaming test on freshly cut stem sections.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes rapid, total systemic wilting and complete plant mortality, resulting in severe crop losses and harvest destruction. Specific numerical impacts on fruit quality and yield loss percentages are not specified in the provided sources.
- **Other Relevant Disease-Specific Information:** Utilizing resistant tomato varieties is a primary defense; resistant and tolerant varieties listed in the sources

include Arka Ananya, Arka Abhijit, Arka Abha, and Arka Alok. Source documents reflect historical and current taxonomic revisions of the causal bacterium, citing it variously as *Ralstonia solanacearum*, *Pseudomonas solanacearum*, and *Burkholderia solanacearum*, and noting five distinct races.

1.5 Tomato Leaf Curl Disease

- **Disease Name and Causal Organism:** Tomato Leaf Curl Disease, caused by Tomato Leaf Curl Virus (ToLCV/TLCV). The causal pathogen consists of isometric virus particles measuring approximately 70 to 90 nm in diameter (with an average reported diameter of 80 nm) and possesses a thermal inactivation point of 40 °C.
- **Symptoms and Signs:** The disease is characterized by severe stunting of plants accompanied by downward rolling and crinkling of leaves. Newly emerging leaves exhibit a slight yellow coloration before curling, whereas older leaves become leathery and brittle. Node and internode lengths are significantly reduced, giving infected plants a pale, bushy appearance with excessive lateral branching. Additional foliage signs include chlorotic spots, general leaf yellowing, and eventual drying of leaves.
- **Disease Development and Spread:** Primary transmission occurs via the whitefly vector (*Bemisia tabaci*) and through plant grafting. A single viruliferous whitefly can effectively transmit the virus. The disease is not sap-transmissible, though seeds harvested from infected fruits may carry viral particles on their outer surface. Whiteflies acquire the virus while feeding on infected tomato plants or solanaceous weed hosts and persistently transmit it to healthy crops during feeding. Secondary spread within the field occurs rapidly as whitefly populations build up.
- **Conditions Favourable for Disease Development:** Warm weather conditions that encourage the rapid multiplication and high population density of the whitefly vector (*Bemisia tabaci*), along with the presence of infected alternate or collateral solanaceous weed hosts harboring both the virus and vector.
- **Diagnosis and Identification:** Identified visually in the field through symptom evaluation, including severe plant stunting, shortened internodes, bushy growth habit, leaf crinkling, chlorosis, and downward leaf curling. Specific laboratory diagnostic and identification procedures are not specified in the provided sources.
- **Prevention and Cultural Practices:** Select seeds exclusively from disease-free, healthy plants. Raise seedlings in protected nursery beds covered with 100-mesh nylon nets, net houses, or greenhouses to exclude whiteflies. Rogue and destroy infected seedlings and plants in both nursery and main field stages; do not transplant diseased seedlings. Eradicate alternate solanaceous weed hosts during weeding and earthing-up operations. Establish tall cereal guard/barrier crops (such as 4 to 6 rows of maize or sorghum) around the tomato field prior to planting. Plant repellant crops such as peppermint or *Ocimum*/basil. Practice crop rotation with non-host crops, including cereals, legumes, cucurbits, or cruciferous vegetables. Cultivate tolerant or resistant varieties/hybrids, such as Arka Ananya, Kashi Vishesh, Kashi Amrit, COTH 2, and TNAU Tomato Hybrid Co3.

- **Biological Control:** Conserve natural nymphal and pupal parasitoids of whiteflies, including *Encarsia* species (*Encarsia formosa*) and *Eretmocerus* species. Protect natural predators such as mirid bugs (*Dicyphus hesperus*), dragonflies, spiders, robber flies, praying mantises, fire ants, coccinellids (ladybird beetles), lacewings, and big-eyed bugs (*Geocoris* species). Apply neem-based biopesticides, including Neem Seed Kernel Extract (NSKE) 5% or Azadirachtin 5% W/W neem extract concentrate @ 80 g in 160 l of water per acre, or Azadirachtin 1% (10000 ppm) @ 400–600 ml in 200 l of water per acre.

- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Dip seedling roots prior to transplanting for 15 minutes in Imidacloprid 17.8% SL @ 60–70 ml in 200 l of water per acre to manage whitefly vectors. For chemical vector control in the main field, apply foliar sprays of Imidacloprid 17.8% SL @ 60–70 ml in 200 l of water per acre, Dimethoate 30% EC @ 396 ml in 200–400 l of water per acre (or Dimethoate 0.05% sprayed at 15, 25, and 45 days after transplanting), Thiamethoxam 25% WG @ 80 g in 200 l of water per acre, Carbofuran 3% CG @ 16,000 g per acre, Malathion 50% EC @ 600 ml in 200–400 l of water per acre, Oxydemeton-methyl 25% EC @ 400 ml in 200–400 l of water per acre, Phorate 10% CG @ 6,000 g per acre, or Spiromesifen 22.9% SC @ 250 ml in 200 l of water per acre.

- **Integrated Disease Management:** Integrates protected nursery seedling production under 100-mesh netting, healthy seed selection, roguing of infected plants, weed sanitation, planting tall cereal barrier crops and repellant plants, setting up yellow sticky traps, conserving biological control agents, applying neem extracts, seedling root dipping, and applying need-based chemical insecticide sprays targeting whitefly vectors.

- **Monitoring and Early Detection:** Install yellow sticky traps or cards at a density of 10 to 12 per acre (or 12 per hectare) across nurseries and main fields to monitor whitefly population dynamics and reduce vector density. Regularly inspect young plants for early symptoms, such as slight yellowing and marginal leaf curling on new foliage.

- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes severe plant stunting, marked reduction in internode lengths, pale bushy growth habit, and severe disruption of flowering and fruit set. Severely infected plants bear very few or no marketable fruits, leading to substantial loss in total crop yield.

- **Other Relevant Disease-Specific Information:** The sources indicate a minor nuance regarding seed transmission, noting that ToLCV is not internally seed-transmitted, though viral particles may be carried externally on seeds harvested from infected fruits, making seed selection from healthy plants essential. Additionally, vector spray recommendations vary between fixed post-transplanting interval schedules (15, 25, and 45 days) and need-based applications depending on whitefly field pressure.

1.6 Damping-Off

- **Disease Name and Causal Organism:** Damping-Off, caused by soil-borne fungal and oomycete pathogens including *Pythium aphanidermatum*, *Pythium* species, *Phytophthora* species, and *Rhizoctonia solani* (teleomorph: *Thanatephorus cucumeris*).

- **Symptoms and Signs:** The disease manifests in two distinct phases: pre-emergence and post-emergence. In the pre-emergence phase, seeds rot before germinating or young seedlings are killed prior to breaking through the soil surface; the radicle and plumule rot completely, giving the appearance of poor germination. *Pythium* and *Phytophthora* species cause dark brown to black water-soaked lesions and soft mushy rot, whereas *Rhizoctonia solani* produces tan to reddish-brown lesions and death of the growing tip. In the post-emergence phase, infection attacks the juvenile collar tissues at the soil level. The infected stem tissues become soft, water-soaked, dark brown to black, or develop constricted reddish-brown lesions. The lesions girdle the stem base, causing seedlings to wilt, collapse, and topple over. Older plants infected by *R. solani* show increased tolerance, with lesions typically restricted to the cortical zone.

- **Disease Development and Spread:** Pathogens persist in the soil for long periods in plant debris, on weed roots, or on alternate hosts. Primary spread occurs via contaminated soil, infected seeds, and water. Secondary spread occurs through conidia dispersed by wind, rain splashes, and water splashing infested soil from diseased to healthy seedlings.

- **Conditions Favourable for Disease Development:** High humidity, excessive soil moisture, poorly drained fields, cloudiness, low temperatures (below 24 °C for a few days), crowded seedling populations, soil compaction, poor ventilation, excess soil solutes, and the use of improperly pasteurized soil or unsterilized seedling trays.

- **Diagnosis and Identification:** Identified visually in nursery beds or seedling trays by gaps in seedling emergence (pre-emergence) and the presence of soft, water-soaked, or constricted stem bases that cause young seedlings to collapse and topple over. Specific laboratory diagnostic methods are not specified in the provided sources.

- **Prevention and Cultural Practices:** Raise seedlings on 15 cm raised seed beds or in pro-trays/sterilized planting trays to ensure proper drainage and aeration. Perform soil solarization by covering nursery beds with 45-gauge (0.45 mm) polythene sheets for three weeks prior to sowing. Practice deep summer ploughing, field sanitation, and crop rotation with non-host cereal crops. Avoid overwatering, poorly drained areas, seedling overcrowding, and the transfer of weed seedlings. Maintain good greenhouse sanitation, proper soil pasteurization, and soil fumigation or solarization in fields.

- **Biological Control:** Treat seeds with *Trichoderma viride* (4 g/kg of seed or 1% WP @ 9 g/kg of seed) or *Pseudomonas fluorescens*. Apply *Trichoderma* and *Pseudomonas fluorescens* as nursery treatments and soil applications.

- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Treat seeds before sowing with chemical fungicides such as Thiram @ 3 g/kg of seed or Captan 75% WS @ 20–30 g/kg of seed. Drench infected seedling beds or soil using Copper oxychloride (0.2%), Bordeaux mixture (1%), Dithane M-45 (3 g/l of water), or Captan 75% WP @ 1000 g in 400 l of water per acre. During cloudy weather, apply foliar sprays of Metalaxyl (0.2%) to control post-emergence damping-off.
- **Integrated Disease Management:** Combines soil solarization, raised seed beds or pro-trays, regulated irrigation, seed treatments with biological cultures (*Trichoderma viride*, *Pseudomonas fluorescens*) or chemical fungicides (Thiram, Captan), and targeted soil drenching (Copper oxychloride, Bordeaux mixture, Dithane M-45, Captan) or Metalaxyl sprays under cloudy conditions.
- **Monitoring and Early Detection:** Regularly inspect nursery beds and seedling trays for uneven germination or post-emergence water-soaking and seedling collapse at the soil line.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes severe seedling mortality in the nursery, complete stand failure in severe pre-emergence cases, and a shortage of healthy transplants for field establishment. Specific effects on fruit quality and mature crop yield are not specified in the provided sources beyond nursery seedling losses.
- **Other Relevant Disease-Specific Information:** The sources contain variable recommendations regarding seed treatment dosages for *Trichoderma viride* (4 g/kg of seed versus 1% WP @ 9 g/kg of seed) and chemical options (Thiram @ 3 g/kg versus Captan 75% WS @ 20–30 g/kg). Soil drench recommendations also vary among Copper oxychloride (0.2%), Bordeaux mixture (1%), Dithane M-45 (3 g/l), and Captan 75% WP (1000 g in 400 l water/acre).

1.7 Tomato Spotted Wilt Disease

- **Disease Name and Causal Organism:** Tomato Spotted Wilt Disease (also known as Tomato Spotted Wilt), caused by Tomato Spotted Wilt Virus (TSWV), Peanut Bud Necrosis Virus (PbNV), or Groundnut Bud Necrosis Virus. The pathogen consists of isometric virus particles measuring 70 to 90 nm in diameter, with a thermal inactivation point of 40 °C. There are multiple viral strains.
- **Symptoms and Signs:** Foliar symptoms begin as orangish-yellow flecks on older leaves, progressing to small, dark, circular necrotic spots on young foliage. Leaves develop a distinct bronzed appearance, thick veins, and eventually turn dark brown and wither. Dark, shiny streaks appear on terminal branches and stems, and growing tips often undergo dieback. Infected plants exhibit severe stunting, a one-sided growth habit, or general wilting with drooping yellowish foliage. Fruits develop slightly raised green, yellow, or red spots approximately half an inch (0.5 inch) in diameter with characteristic concentric circular markings; ripe fruits show alternate bands or concentric rings of red and yellow or white, fruit deformation, or severe fruit necrosis. Seed discoloration is also produced.

- **Disease Development and Spread:** Primary survival and inoculum sources consist of infected alternate and collateral crop, ornamental, and weed hosts (including *Acanthospermum hispidum*, *Aster* species, *Boerhaavia diffusa*, *Chrysanthemum* species, *Cleome gynandra*, cowpea, *Dahlia variabilis*, eggplant, French bean, *Gerbera* species, groundnut, *Lagasca mollis*, lettuce, marigold, pea, chili, pineapple, potato, *Trianthema portulacastrum*, watermelon, and *Zinnia elegans*). Secondary transmission occurs persistently via thrips vectors, specifically *Thrips tabaci*, *Frankliniella schultzei*, *Frankliniella occidentalis*, and *Scirtothrips dorsalis*. Thrips acquire the virus during larval feeding, and winged adult thrips are blown by wind into tomato crops from surrounding infected weeds or ornamentals. Spread directly from tomato plant to tomato plant within the field is typically minimal.

- **Conditions Favourable for Disease Development:** High population densities of thrips vectors, abundance of infected alternate or collateral weed and ornamental host plants in proximity to tomato fields, and wind movement facilitating adult thrips migration into the crop.

- **Diagnosis and Identification:** Identified visually in the field by leaf bronzing, dark circular leaf spots, terminal dieback, stem streaking, stunted or one-sided plant growth with drooping foliage, and ripe fruits displaying alternate red and yellow concentric rings. Specific laboratory diagnostic methods are not specified in the provided sources.

- **Prevention and Cultural Practices:** Systematically remove and destroy diseased plants from tomato fields. Eradicate alternate or collateral weed and ornamental hosts in surrounding areas during weeding and earthing-up operations. Establish 5 to 6 rows of tall barrier crops (such as sorghum, maize, or bajra) around the tomato field prior to planting to block incoming viruliferous thrips. Cultivate resistant tomato varieties or cultivars.

- **Biological Control:** Not specified in the provided sources.

- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Apply foliar sprays of systemic insecticides to control thrips vector populations, specifically Imidacloprid (0.05%), or apply 2 to 3 foliar sprays of Dimethoate (0.05%) or Monocrotophos (0.05%) at 10-day intervals.

- **Integrated Disease Management:** Combines resistant variety selection, roguing of infected crop plants, weed sanitation around fields, planting 5 to 6 rows of tall cereal barrier crops (maize, sorghum, or bajra), and targeted foliar application of systemic insecticides (Imidacloprid 0.05%, Dimethoate 0.05%, or Monocrotophos 0.05%) to reduce thrips vector populations.

- **Monitoring and Early Detection:** Inspect young foliage regularly for early symptoms such as orangish-yellow flecks, dark circular spots, leaf bronzing, and check surrounding weeds and tomato plants for thrips activity.

- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes severe plant stunting, dieback of growing tips, one-sided growth habits, fruit deformation, unmarketable fruit with red-and-yellow concentric ring blemishes, and seed discoloration, leading to substantial reductions in total crop yield and commercial market value.

- **Other Relevant Disease-Specific Information:** A key disease-specific mechanism is that thrips acquire the virus exclusively during their larval stage, but only adult thrips transmit the virus when feeding on tomato crops. Consequently, initial disease development in a field is predominantly driven by wind-blown adult thrips migrating from adjacent weed hosts rather than secondary spread within the tomato planting.

1.8 Root-Knot Nematode

- **Disease Name and Causal Organism:** Root-Knot Nematode (synonyms: Root-Knot or Root Gall), caused by microscopic plant-parasitic nematodes of the genus *Meloidogyne*, including *Meloidogyne incognita*, *Meloidogyne hapla*, *Meloidogyne javanica*, and *Meloidogyne arenaria*.

- **Symptoms and Signs:** Primary symptoms consist of galls or irregular swellings and knots on the host root system. Roots branch profusely starting from gall tissue, producing a "beard root" appearance; severely infected roots become knobby, with rootlets almost completely absent. Above-ground symptoms include patchiness in the field, general lack of plant vigor, stunting, chlorosis, wilting during hot parts of the day (especially under dry conditions), progressive drying of older leaves, and a purple discoloration on the undersides of leaves resembling phosphate deficiency.

- **Disease Development and Spread:** The life cycle comprises eggs, four juvenile/larval stages, and adults (longer males lasting 16–22 days and bulged females lasting 25–30 days). First-stage larvae moult inside eggs, and infective second-stage juveniles (J2) hatch in 4–6 days to invade host roots. The complete life cycle takes 4 to 8 weeks depending on temperature. Primary survival and inoculum sources include cysts and egg masses in infected plant debris, infested soil, weeds, or alternate crops (solanaceous, malvaceous, and leguminaceous plants), as well as contaminated transplants, equipment, or irrigation water. Secondary spread occurs through autonomous movement of second-stage juveniles and water dispersion.

- **Conditions Favourable for Disease Development:** Light, sandy, or loamy soils, long growing seasons with mild winters, and optimal soil temperatures ranging from 70 to 80°F (16–20°C for *M. hapla* and 27°C / 81°F for *M. incognita*).

- **Diagnosis and Identification:** Identified visually in the field by the presence of stunted or wilting plants appearing in patches, followed by uprooting plants to inspect for root galls, knobby roots, and excessive "beard root" branching. Specific laboratory diagnostic procedures are not specified in the provided sources.

- **Prevention and Cultural Practices:** Deep summer ploughing to expose nematodes to heat. Soil solarization of nursery beds with 45-gauge (0.45 mm) polythene sheets for three weeks prior to sowing, or soil pasteurization and fumigation. Raise seedlings in nematode-free sites or solarized beds, and use certified disease-free transplants. Practice crop rotation with non-host cereal crops or rotation crops such as marigold, *Chrysanthemum* species, *Sesbania* species, *Crotalaria* species, *Gaillardia* species, castor bean, and *Desmodium* species. Intercrop or use ecological engineering with tall marigold as a trap crop (@ 16:1 ratio, transplanting 45-day-old marigold with 25-day-old tomato). Apply neem cake @ 500 kg/ha (or

100 kg/acre) to the main field prior to transplanting. Plant resistant or tolerant varieties/hybrids, such as Pusa-120, Pusa Hybrid-2, Pusa Hybrid-4, Arka Vardan, Hisar Lalit, and TNAU Tomato Hybrid Co3.

- **Biological Control:** Treat seeds with bio-control agents using IIHR organic formulation (*Pseudomonas fluorescens* + *Trichoderma harzianum*) @ 15–20 g/kg of seed, *Trichoderma viride* or *P. fluorescens* @ 200 g/kg of seed, or *Trichoderma viride* 1% WP @ 9 g/kg of seed. Dip seedling roots in bio-control solutions (*P. fluorescens* + *T. harzianum* @ 5 g/l or *T. viride* / *P. fluorescens* @ 5 g/l or 5 ml/l for 10–15 minutes). Apply farmyard manure @ 2 t/ha enriched with *T. viride* or *P. fluorescens* @ 2 kg/t, or vermicompost @ 1 t/ha. Apply egg-parasitic biocontrol agents such as *Paecilomyces lilacinus*.

- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Perform seedling root dipping in an insecticide plus fungicide combination (Thiamethoxam 25 WG + Captan 75% WP @ 1 g/l + Copper oxychloride @ 2.5 g/l) for 10–15 minutes prior to transplanting. Apply Dazomet technical @ 12–16 g/acre to nursery beds. Perform soil fumigation or pasteurization where feasible.

- **Integrated Disease Management:** Integrates deep summer ploughing, soil solarization, raising seedlings in nematode-free or solarized beds, cultivating resistant varieties (e.g., Arka Vardan, Hisar Lalit), crop rotation with cereals or non-hosts, intercropping with marigold trap crops, applying neem cake and enriched farmyard manure, seed and seedling root treatment with bio-agents (*Trichoderma*, *Pseudomonas*, *Paecilomyces lilacinus*), and nursery treatment with Dazomet technical.

- **Monitoring and Early Detection:** Regularly monitor fields for patches of stunted, off-colored, or wilting plants during dry or hot periods, and inspect seedling roots in nurseries for early galling prior to field transplanting.

- **Effects on Crop Growth, Fruit Quality, and Yield:** Disrupts root function in absorbing and transporting water and nutrients, leading to plant stunting, wilting, chlorosis, and severe reduction in flowering and fruit set. Seedlings infected in the nursery rarely survive transplanting. Specific effects on fruit quality are not specified in the provided sources beyond reduced fruit production.

- **Other Relevant Disease-Specific Information:** Nematode root damage predisposes tomato plants to secondary infection by fungal and bacterial root pathogens, and exacerbates physiological disorders such as blossom-end rot by impairing calcium and water uptake. Varietal resistance varies by species; commercial varieties resistant to *M. hapla* are currently unavailable, whereas resistant varieties exist for *M. incognita*. Furthermore, seed treatment dosage recommendations for biological control agents vary in the sources between 15–20 g/kg (formulated mixture) and 200 g/kg of seed.

1.9 Bacterial Spot

- **Disease Name and Causal Organism:** Bacterial Spot (also known as Bacterial Leaf Spot), caused by *Xanthomonas campestris* pv. *vesicatoria* (synonyms: *Xanthomonas vesicatoria*, *Xanthomonas euvesicatoria*, *Xanthomonas perforans*, and

Xanthomonas gardneri). The pathogen is a gram-negative, short rod-shaped bacterium possessing a single polar flagellum and a protective capsule. Five distinct races of the bacterium have been reported.

- **Symptoms and Signs:** The disease affects all above-ground plant parts, including leaves, stems, petioles, and green fruits. Foliar symptoms begin as small (less than 3 mm in diameter), dark brown to black, water-soaked circular spots surrounded by a prominent yellowish halo; spot centers dry out and crack, acquiring an angular appearance with a translucent center and dark border. On older plants, infection primarily targets lower leaves, causing severe leaf blighting and defoliation. The most striking symptoms occur on green fruits, initiating as small water-soaked spots that become raised and enlarge to one-eighth to one-fourth inch (4–5 mm) in diameter; lesion centers turn light brown to black, slightly sunken or raised, with a rough, scabby surface and occasional white halos or raised margins.

- **Disease Development and Spread:** Primary sources of survival and inoculum include infected plant debris in the soil, volunteer tomato plants, alternate solanaceous weed hosts (such as *Solanum nigrum*), and contaminated seed surfaces or internal seed tissues. Bacteria enter host tissues through stomata, lenticels, or mechanical wounds caused by insects, wind-driven sand/rain, or high-pressure spraying. Secondary spread within nurseries and main fields occurs rapidly via rain splashes, wind-driven rainstorms, splattering irrigation water, infected seeds, and diseased transplants.

- **Conditions Favourable for Disease Development:** Warm weather conditions with temperatures between 24–30 °C (75–86 °F), high atmospheric humidity, persistent dew, fog, splattering rains, heavy rainstorms, and overhead sprinkler irrigation.

- **Diagnosis and Identification:** Identified visually in the field by small water-soaked leaf spots with distinct yellow halos, extensive lower leaf defoliation, and characteristic raised, rough, scabby lesions (1/8 to 1/4 inch) on green fruits. Specific laboratory diagnostic procedures are not specified in the provided sources.

- **Prevention and Cultural Practices:** Select disease-free certified seeds and transplants. Practice crop rotation with non-host plant species (such as cereals) to avoid exposure to infected crop residues from previous seasons. Implement thorough field sanitation by cleaning machinery used in infected fields, plowing under crop debris immediately after harvest, and eradicating alternate solanaceous weeds and volunteer tomato plants. Avoid overhead sprinkler irrigation in favor of drip or furrow irrigation to minimize foliage wetness, and avoid mechanical wounding of plants during field operations.

- **Biological Control:** Not specified in the provided sources.

- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Perform seed treatment with mercuric chloride (1:1000) or streptocycline. Spray streptomycin sulfate 9% + tetracycline hydrochloride 1% SP solution (streptocycline) @ 40–100 ppm twice (one application in nursery seedbeds before transplanting and a second in the main field after transplanting). Dip seedling roots prior to transplanting in an insecticide-fungicide combination (Thiamethoxam 25 WG +

Captan 75% WP @ 1 g/l + Copper oxychloride @ 2.5 g/l). In the main field, apply foliar sprays of copper-based fungicides (such as Copper oxychloride 50% WP @ 2–3 g/l or 1000 g in 300–400 l of water per acre) combined with organic fungicides at 5 to 10-day intervals, or apply Agrimycin-100 (100 ppm) three times at 10-day intervals.

- **Integrated Disease Management:** Integrates disease-free seed selection, seed treatment (mercuric chloride 1:1000 or streptocycline @ 40–100 ppm), seedling root dips, non-host crop rotation, field sanitation (weed and plant debris destruction), avoidance of overhead irrigation, and regular preventive foliar sprays combining copper fungicides (Copper oxychloride) with organic fungicides or bactericidal applications (Agrimycin-100).
- **Monitoring and Early Detection:** Monitor seedbeds and main fields closely after rainstorms or during extended periods of humid, wet weather; inspect lower leaves for water-soaked spots with yellow halos and green fruits for early raised lesions.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes severe defoliation of foliage, reduced photosynthetic capacity, loss of plant vigor, and unsightliness of green fruits due to scabby lesions, rendering fruit unmarketable and leading to substantial yield and economic losses.
- **Other Relevant Disease-Specific Information:** Ripe fruits are not susceptible to direct infection, but the bacterial pathogen readily contaminates the outer surface of seeds harvested from infected fruits, remaining viable externally for extended periods.

1.10 Septoria Leaf Spot

- **Disease Name and Causal Organism:** Septoria Leaf Spot, caused by the fungus *Septoria lycopersici*. At least two distinct races of the pathogen have been reported.
- **Symptoms and Signs:** Infection can occur at any growth stage, typically appearing first on the older, lower leaves of less vigorous plants. Initial foliage symptoms present as small, dark, water-soaked circular lesions. These enlarge to approximately 5 mm (0.2 inches) in diameter, developing gray to brownish-gray centers peppered with tiny, black, pimple-like fruiting bodies (pycnidia), surrounded by a distinct dark brown or black margin and a chlorotic yellow area. As spots coalesce, leaves dry up, collapse, turn blighted, and undergo severe defoliation, progressing upward from the bottom of the plant. Elongated lesions containing pycnidia in their centers can also develop on stems, petioles, and calyxes, and flowers are occasionally attacked. Fruits are rarely infected directly.
- **Disease Development and Spread:** The pathogen overwinters and survives primarily as mycelium or conidia in pycnidia within infected crop debris in or on the soil for one or more years, as well as on alternate solanaceous weeds (such as nightshade, horse nettle, jimson weed, and ground cherry) and volunteer tomato plants. The fungus can also be seed-borne, resulting in infected seedlings. Secondary spread occurs when fungal conidia are exuded from pycnidia under high

humidity (100%) and dispersed by rain splashes, wind, overhead sprinkler irrigation, cultivation equipment, insects, and slimy conidia adhering to the hands and clothing of workers during harvesting.

- **Conditions Favourable for Disease Development:** Extended periods of high relative humidity (100% or persistent dew), temperatures between 20–25 °C (68–77 °F), moist weather with intermittent rain showers, poor plant vigour resulting from nutrient inadequacy or late-season crop growth, and the use of overhead sprinkler irrigation.
- **Diagnosis and Identification:** Identified visually in the field by small (5 mm) circular spots on lower foliage possessing gray centres peppered with black pycnidia, dark borders, and yellow halos, accompanied by upward disease progression and lower leaf defoliation. Specific laboratory diagnostic procedures are not specified in the provided sources.
- **Prevention and Cultural Practices:** Collect, remove, and destroy or turn under infected plant parts and crop debris after harvest. Implement a three-year crop rotation with non-host crops. Maintain adequate soil fertility and proper fertilization to preserve plant vigor. Eradicate volunteer tomato plants and solanaceous weeds (nightshade, horse nettle, jimson weed, and ground cherry) in and around tomato fields. Avoid overhead sprinkler irrigation to keep foliage dry, and refrain from handling or harvesting wet plants to prevent spreading conidia.
- **Biological Control:** Not specified in the provided sources.
- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Perform seed treatment prior to sowing using Thiram or Dithane M-45 @ 2 g/kg of seed to check seed-borne infection. For field control, apply foliar sprays of Mancozeb 75% WP @ 600–800 g in 300 l of water per acre (or Mancozeb 0.2%), Dithane Z-78 (0.2%), Dithane M-45 (0.2%), Zineb (0.2%), or Copper oxychloride (0.2%) at 10 to 14-day intervals upon disease appearance.
- **Integrated Disease Management:** Combines seed treatment (Thiram or Dithane M-45 @ 2 g/kg seed), proper plant nutrition to maintain vigor, destruction of crop debris and alternate solanaceous weeds, a three-year crop rotation, avoidance of overhead sprinkler irrigation, and timely preventive foliar sprays of fungicides such as Mancozeb 75% WP (@ 600–800 g in 300 l of water per acre or 0.2%) or Dithane Z-78 / Dithane M-45 (0.2%).
- **Monitoring and Early Detection:** Regularly inspect the lower leaves of tomato plants, particularly during periods of warm, wet weather, high humidity, or nutrient stress, for initial small water-soaked spots with gray centers and tiny black pycnidia.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes extensive leaf blighting and severe lower leaf defoliation, markedly reducing plant photosynthetic capacity, overall vigor, and total fruit yield. While fruits are rarely attacked directly, extensive loss of foliage exposes developing fruits to sunscald and reduces fruit production.

- **Other Relevant Disease-Specific Information:** Conidia of *Septoria lycopersici* are exuded in a gelatinous, slimy mass from pycnidia under wet conditions, allowing them to readily stick to workers' hands, tools, and clothing during field operations, making field sanitation and dry foliage management during picking particularly critical.

1.11 Tomato Yellow Leaf Curl Disease

- **Disease Name and Causal Organism:** Tomato Yellow Leaf Curl Disease (also referred to as Tomato Leaf Curl Disease or Tomato Leaf Curl), caused by Tomato yellow leaf curl virus (TYLCV) and Tomato leaf curl virus (ToLCV or TLCV). The causal geminivirus particles are isometric, measuring approximately 80 nm in diameter (ranging from 70 to 90 nm), with a reported thermal inactivation point of 40 °C.

- **Symptoms and Signs:** Early-stage infection causes severe plant stunting, erect branching, and small chlorotic leaflets that cup and twist upward. Newly emerging leaves exhibit slight yellow coloration before curling downward and crinkling, while older leaves become leathery, brittle, or develop interveinal chlorosis. Nodes and internodes are significantly reduced in size, giving infected plants a pale, bushy appearance with excessive lateral branching. Infection occurring at later growth stages causes leaf cupping, leaflet yellowing, flower abortion, and failure to set fruit. Fruits that set prior to infection generally ripen normally.

- **Disease Development and Spread:** Primary transmission occurs in a persistent manner via whitefly vectors, specifically the sweet potato whitefly or silverleaf whitefly (*Bemisia tabaci* and *Bemisia argentifolii*), as well as through plant grafting. Whitefly larvae acquire the virus while feeding on infected tomato plants or alternate solanaceous weed hosts, and adult whiteflies persistently transmit it to healthy tomato crops. A single viruliferous whitefly can transmit the virus. Secondary spread within the field occurs rapidly as vector populations build up. The virus is not mechanically transmitted (not sap-transmissible) and cannot be spread from plant to plant by workers or equipment, though seeds harvested from infected fruits may carry viral particles on their outer surface.

- **Conditions Favourable for Disease Development:** Warm weather conditions encouraging rapid vector multiplication, high population densities of whiteflies (*Bemisia tabaci / Bemisia argentifolii*), and the presence of infected alternate or collateral solanaceous weed hosts harboring both the virus and vector.

- **Diagnosis and Identification:** Identified visually in the field through symptom evaluation, including severe plant stunting, erect or bushy growth habit, shortened internodes, leaf cupping, upward twisting, downward leaf rolling, crinkling, interveinal chlorosis, and flower abortion. Specific laboratory diagnostic procedures are not specified in the provided sources.

- **Prevention and Cultural Practices:** Cultivate resistant or tolerant varieties/hybrids, such as Arka Ananya, Kashi Vishesh, Kashi Amrit, COTH 2, and TNAU Tomato Hybrid Co3. Select seeds exclusively from disease-free, healthy plants. Raise seedlings in protected nursery beds covered with 100-mesh nylon nets, net houses,

or greenhouses to exclude whiteflies. Carefully rogue and destroy infected seedlings in nurseries and diseased plants in main fields; do not transplant infected seedlings. Eradicate alternate solanaceous weed hosts in and around fields during weeding and earthing-up operations. Adopt crop-free periods where feasible. Establish tall cereal barrier/border crops (such as 5 to 6 rows of maize, sorghum/jowar, or bajra) around tomato fields at least 50–60 days prior to transplanting to check incoming viruliferous whiteflies. Plant repellant crops such as peppermint, marigold, or *Ocimum*/basil. Cover plant beds with yellow plastic mulch (which attracts whiteflies) or white, blue, grey, and black Low Density Polyethylene (LDPE) sheets (30 micron) prior to transplanting. Practice crop rotation with non-host crops, including cereals, legumes, cucurbits, or cruciferous vegetables.

- **Biological Control:** Conserve natural whitefly parasitoids, including *Encarsia* species (*Encarsia formosa*) and *Eretmocerus* species (nymphal and pupal stages). Conserve natural predators, such as mirid bugs (*Dicyphus hesperus*), dragonflies, spiders, robber flies, praying mantises, fire ants, coccinellids (ladybird beetles), lacewings, and big-eyed bugs (*Geocoris* species). Apply neem-based biopesticides, including Neem Seed Kernel Extract (NSKE) 5%, Azadirachtin 5% W/W neem extract concentrate @ 80 g in 160 l of water per acre, Azadirachtin 1% (10000 ppm) @ 400–600 ml in 200 l of water per acre, or Neem oil (Azadirachtin 5% EC) @ 2 ml/l.

- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Perform soil application of granular insecticide in nursery beds at sowing using Carbofuran / Furadan @ 1 kg a.i./ha, followed by a second application @ 1.5 kg a.i./ha one week after transplanting. Dip seedling roots prior to transplanting for 15 minutes in Imidacloprid 17.8% SL @ 60–70 ml in 200 l of water per acre to manage vector populations. Apply regular foliar sprays of mineral oil to slow virus spread by reducing whitefly acquisition and transmission. For chemical vector control in the main field, apply foliar sprays of Imidacloprid 17.8% SL @ 60–70 ml in 200 l of water per acre (or 0.05%), Dimethoate 30% EC @ 396 ml in 200–400 l of water per acre (or 0.05% sprayed at 10-day intervals or at 15, 25, and 45 days after transplanting), Monocrotophos (0.05%) at 10-day intervals, Thiamethoxam 25% WG @ 80 g in 200 l of water per acre, Carbofuran 3% CG @ 16,000 g per acre, Malathion 50% EC @ 600 ml in 200–400 l of water per acre, Oxydemeton-methyl 25% EC @ 400 ml in 200–400 l of water per acre, Phorate 10% CG @ 6,000 g per acre, or Spiromesifen 22.9% SC @ 250 ml in 200 l of water per acre.

- **Integrated Disease Management:** Integrates resistant or tolerant varieties (Arka Ananya, Kashi Vishesh, etc.), protected nursery seedling production under 100-mesh netting, healthy seed selection, yellow plastic or LDPE polythene mulching, soil granular insecticide application (Furadan), seedling root dips (Imidacloprid), tall cereal barrier cropping, yellow sticky traps, weed sanitation, mineral oil sprays, biological parasitoid/predator conservation, neem biopesticides, and need-based chemical sprays.

- **Monitoring and Early Detection:** Install yellow sticky traps or cards @ 10 to 12 per acre (or 12 per hectare) placed 15 cm above the canopy across nurseries and main fields to monitor whitefly population dynamics and reduce vector density.

Regularly inspect young plants for early symptoms, such as slight leaf yellowing, erect branching, and upward or downward leaf cupping and curling.

- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes severe plant stunting, marked reduction in internodes, pale bushy growth, flower abortion, and drastic failure to set fruit when infection occurs at early growth stages. Plants infected early bear very few or no marketable fruits, resulting in severe crop yield loss.

- **Other Relevant Disease-Specific Information:** The sources highlight that the virus is strictly vector-transmitted by whiteflies and is not mechanically transmissible by workers or tools. Recommendations vary regarding chemical vector control timing, including fixed post-transplanting spray schedules (15, 25, and 45 days after transplanting) versus 10-day interval sprays, as well as the use of systemic soil granules versus foliar sprays. Additionally, silverleaf whitefly (*Bemisia argentifolii*) nymphal feeding is specifically associated with the physiological fruit disorder known as irregular ripening in tomatoes.

1.12 Peanut Bud Necrosis Disease

- **Disease Name and Causal Organism:** Peanut Bud Necrosis Disease (also referred to as Tomato Spotted Wilt Disease or Tomato Bud Necrosis Disease) is caused by the Peanut Bud Necrosis Virus (PbNV) / Groundnut Bud Necrosis Virus, which is grouped with Tomato Spotted Wilt Virus (TSWV). The virus particles are isometric and range from 70 to 90 nm in diameter, with a thermal inactivation point of 40 °C.

- **Symptoms and Signs:** The disease is characterized by streaking on leaves, stems, and fruits. Young leaves develop numerous small, dark, circular spots, vein thickening, and a bronzed appearance, eventually turning dark brown and withering. Growing tips may die back, terminal branches may show streaking, and affected plants exhibit a one-sided growth habit or complete stunting with drooping leaves. Fruits display spots about half an inch in diameter with concentric circular markings, which appear as alternate bands of red and yellow or pale red and yellow areas on ripe fruit. Discoloration of seeds also occurs.

- **Disease Development and Spread:** Primary survival and spread occur through virus particles in infected host plants, including groundnut, chili, eggplant, potato, pea, French bean, cowpea, watermelon, lettuce, marigold, pineapple, and various weeds and ornamentals such as *Acanthospermum hispidum*, *Aster* sp., *Boerhaavia diffusa*, *Chrysanthemum* sp., *Cleome gynandra*, *Dahlia variabilis*, *Gerbera* sp., *Lagasca mollis*, *Trianthema portulacastrum*, and *Zinnia elegans*. Secondary spread is driven by vector thrips species transmitting the virus particles, including *Thrips tabaci*, *Frankliniella schultzi*, *Frankliniella occidentalis*, and *Scirtothrips dorsalis*.

- **Conditions Favourable for Disease Development:** High populations of thrips vectors and the presence of infected alternate or collateral hosts in and around the cultivated field favor disease outbreak and spread.

- **Diagnosis and Identification:** Identified by distinct visual symptoms including leaf bronzing, small dark circular spots on young foliage, terminal stem streaking, die-back, plant stunting with drooping leaves, and fruits marked with concentric circular rings or red-yellow color bands.
- **Prevention and Cultural Practices:** Promptly remove and destroy infected tomato plants. Eradicate alternate or collateral weed hosts that harbor the virus during field weeding or earthing-up operations. Plant barrier crops such as sorghum, maize, or bajra in 5 to 6 rows around the tomato plot prior to transplanting to restrict vector entry.
- **Biological and Chemical Control:** Biological control measures for this specific virus are not specified in the provided sources. Chemical control focuses on suppressing vector thrips populations using targeted insecticide applications.
- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Apply 2 to 3 foliar sprays of Dimethoate (0.05%) or Monocrotophos (0.05%) at 10-day intervals to control vector thrips populations.
- **Integrated Disease Management:** Combine rogueing and destruction of diseased plants, clean cultivation to eliminate alternate host weeds, planting tall cereal barrier crops around fields, and timely chemical sprays against thrips vectors.
- **Monitoring and Early Detection:** Regularly inspect fields for early symptoms on young foliage, such as dark circular spots and bronzing, and observe thrips vector activity on plants.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes severe plant stunting, die-back of shoot terminals, and foliage drooping. Fruit quality is heavily compromised by circular concentric markings, color banding, and seed discoloration, resulting in substantial yield and economic loss.
- **Other Relevant Disease-Specific Information:** The disease is documented in tomato-producing regions including Himachal Pradesh, Andhra Pradesh, Karnataka, and Tamil Nadu.

1.13 Bacterial Stem and Fruit Canker

- **Disease Name and Causal Organism:** Bacterial Stem and Fruit Canker (also known as Bacterial Canker) is caused by the bacterium *Clavibacter michiganensis* subsp. *michiganensis* (syn. *Clavibacter michiganensis* pv. *michiganensis*).
- **Symptoms and Signs:** Initial symptoms appear as downward turning and wilting of lower leaves, often exhibiting unilateral (one-sided) wilting of leaflets along the midrib or rachis. White blister-like spots form on leaf margins, turning brown with age and coalescing without defoliation, while leaves and petioles remain characteristically attached to the stem. Light-colored streaks extend along stems and petioles, eventually cracking open into longitudinal cankers that exude slimy bacterial ooze in humid weather. Internally, stems show light brown, yellow, or reddish-brown vascular discoloration, progressing to a yellow, mealy, and hollow pith. On green

fruits, small, shallow, water-soaked spots develop with distinctive white halos, giving a "bird's-eye" appearance; spot centers become raised, tan-colored, and scabby, accompanied by internal yellow-brown vascular discoloration extending from the stem scar into the fruit pith.

- **Disease Development and Spread:** The bacterium survives primarily in infected plant debris, internal and external seed contamination, volunteer tomato plants, solanaceous weeds such as *Solanum nigrum*, and in soil for up to five years. Secondary spread occurs rapidly through rain splash, splashing water, contaminated clipping or pruning tools, transplanting flats, and worker handling. Secondary spread develops rapidly if transplants are topped or mowed. Infection enters plants through wounds, stomata, or root systems.
- **Conditions Favourable for Disease Development:** Disease development is favoured by temperatures between 18–24 °C (65–75 °F), soil temperatures near 28 °C, high humidity (greater than 80% relative humidity), persistent dew, and moist weather with rain splash. High soil nitrogen levels, low light intensity, and sandy soils further enhance disease severity.
- **Diagnosis and Identification:** Diagnostic features include unilateral wilting with attached leaves, vascular stem discoloration showing a creamy white, yellow, or brown phloem line that exudes bacterial ooze when cut or squeezed, open stem cankers, and fruit spots surrounded by distinctive white halos ("bird's-eye" lesions).
- **Prevention and Cultural Practices:** Cultural preventive measures include using clean, certified disease-free seeds and transplants. Soil, transplanting mix, flats, and field equipment must be thoroughly sanitized and sterilized. Mowing or topping transplants must be strictly avoided to prevent rapid secondary transmission. Crop rotation with non-host crops for at least three years reduces soil inoculum.
- **Biological and Chemical Control:** Biological control options are not specified in the provided sources. Chemical control for seed treatment includes soaking seeds in a Streptocycline solution (1 g per 40 liters of water) for 30 minutes, or seed treatment with mercuric chloride at a 1:1000 ratio.
- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Documented treatments focus on seed sanitation: hot water seed treatment at 50 °C for 25 minutes, seed treatment with mercuric chloride at a 1:1000 ratio, and Streptocycline seed soaking at 1 g per 40 litres of water for 30 minutes to protect early seedling growth. Field chemical spray treatments, doses, and application methods for mature plants are not specified in the provided sources.
- **Integrated Disease Management:** Integrated management involves combining certified pathogen-free seeds, hot water or chemical seed treatments, sterilization of transplant mix and tools, strict avoidance of plant topping or clipping, and maintaining a non-host crop rotation of at least three years.
- **Monitoring and Early Detection:** Not specified in the provided sources.
- **Effects on Crop Growth, Fruit Quality, and Yield:** The disease causes wilting, severe stunting, stem breakdown, canker formation, and death of plants. Fruits

develop bird's-eye blemishes, scabby raised centers, and internal pith discoloration, resulting in unmarketable produce and significant harvest losses.

- **Other Relevant Disease-Specific Information:** Bacterial Stem and Fruit Canker has a worldwide distribution and is recorded in Maharashtra and other tomato-growing states in India.

1.14 Buckeye Rot

- **Disease Name and Causal Organism:** Buckeye Rot (also known as Buckeye Fruit Rot, Buckeye Fruit and Root Rot, or Phytophthora Root Rot) is caused primarily by *Phytophthora parasitica* (syn. *Phytophthora nicotianae* var. *parasitica*), as well as *Phytophthora capsici* and *Phytophthora dreshsleri*.

- **Symptoms and Signs:** The disease primarily attacks fruits near ground level or where they touch the soil, producing grayish-green or brown water-soaked lesions. As lesions enlarge, they form a characteristic pattern of concentric rings consisting of alternating narrow dark brown and wide light brown bands. Young green fruits often become mummified, while mature fruits rot rapidly due to secondary organisms. The pathogen can also cause seedling damping-off, root and crown rot, and stem lesions. Infected taproots and secondary roots collapse with sunken brown lesions and display chocolate-brown vascular discoloration extending beyond the lesion. Foliar symptoms, when present, appear as water-soaked irregular lesions that quickly collapse and dry, leading to wilting and plant death.

- **Disease Development and Spread:** The causal pathogens persist in the soil and infected plant debris for at least two to three years. Primary infection occurs when fruit or lower plant tissue comes into contact with infested soil or irrigation runoff. Secondary spread occurs through rain splash, surface water movement, farm equipment, and irrigation water.

- **Conditions Favourable for Disease Development:** Disease development is favoured by warm soil temperatures around 20 °C (68 °F), high soil moisture, heavy or compacted soils, excessive rainfall or overhead irrigation, and prolonged periods of soil saturation.

- **Diagnosis and Identification:** Diagnosed by the distinct "buckeye" pattern on fruit lesions consisting of smooth concentric rings with alternating dark and light brown bands. Longitudinal sectioning of infected taproots reveals chocolate-brown vascular discoloration. Symptoms are predominantly centred on fruits touching or close to the soil surface.

- **Prevention and Cultural Practices:** Cultural practices include planting in well-drained soils, utilizing raised beds, avoiding soil compaction, applying mulches, and practicing a three-year crop rotation with non-host crops. Staking plants and pruning or clipping lower leaves and fruits up to 15–30 cm from the ground prevents soil contact and splash transmission. Shorter irrigation cycles should be used to avoid extended periods of soil saturation.

- **Biological and Chemical Control:** Biological control involves seed treatment using *Trichoderma* spp. (*Trichoderma viride*). Chemical control includes protective foliar sprays with fungicides such as Difolatan, Mancozeb, and Propineb.
- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Seed treatment with *Trichoderma* spp. protects young seedlings. Recommended field spray treatments include Difolatan (0.3%) applied 4 times at 10-day intervals, Mancozeb 75% WP applied at 600–800 g in 300 liters of water per acre, or Propineb 70% WP applied at 120 g in 40 liters of water per acre.
- **Integrated Disease Management:** Integrated management combines the use of resistant cultivars, biological seed treatments (*Trichoderma* spp.), raised beds, improved soil drainage, soil mulching, plant staking, periodic pruning of lower leaves and fruit up to 30 cm, a three-year non-host crop rotation, and timely chemical fungicide applications (Mancozeb, Propineb, or Difolatan).
- **Monitoring and Early Detection:** Not specified in the provided sources.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Infection leads to seedling mortality, severe stunting, root decay, and plant death. Affected fruits develop severe concentric rot, mummification, or rapid secondary soft rot, causing unmarketable produce and significant yield reductions.
- **Other Relevant Disease-Specific Information:** The provided sources present a slight conflict regarding foliage involvement: one source states that the pathogen affects only fruits near the ground and does not affect foliage (distinguishing it from late blight), whereas another source indicates that the pathogens can infect all plant parts, including leaves, stems, and roots causing foliar blight.

1.15 Gray Mould

- **Disease Name and Causal Organism**: Gray Mould (also spelled Grey Mold), caused by the fungus *Botrytis cinerea* (teleomorph: *Botryotinia fuckeliana*).
- **Symptoms and Signs**: The fungus can infect all above-ground parts of the plant, including stems, leaves, petioles, calyx, and fruits, typically entering through wounds. On stems, initial infection appears as elliptical, water-soaked lesions that, under humid conditions, develop a gray moldy growth that can girdle and kill the plant; stem lesions frequently display concentric banding. On leaves, infection usually starts at injury points and forms a V-shaped lesion covered with gray fungal sporulation. On fruits, the disease commonly attacks the calyx end and spreads rapidly into a watery rot. Fruit lesions initially appear as watery spots with a light brown or tan center, quickly progressing to a soft, mushy mass that breaks down the fruit skin and produces grayish mycelium and spore clusters. Entry points on young green fruits may form a white ring or halo. Exposure to direct sunlight and high temperatures halts fungal progression on green or red fruits, resulting in "ghost spots" (single rings or solid white circular areas with dark brown specks) which do not develop further but reduce market quality.
- **Disease Development and Spread**: *Botrytis cinerea* is an efficient saprophyte with a wide host range. It survives for long periods in soil and infected plant debris

in the form of sclerotia. When adequate humidity is present, masses of gray fungal spores (conidia) are produced and readily disseminated by wind.

- **Conditions Favourable for Disease Development:** High humidity is essential for spore production. Optimal infection temperatures range from 65 °F to 75 °F (18 °C to 24 °C), with infections occurring in as little as 5 hours under favourable conditions. Overcast, cool, and humid weather, along with close plant spacing and poor ventilation, leads to severe gray mould outbreaks. Fungal growth and spore production are inhibited at temperatures exceeding 82 °F (28 °C.
- **Diagnosis and Identification:** Diagnostic signs include soft, mushy fruit decay covered with gray mycelium and spore clusters; elliptical stem lesions with concentric banding; V-shaped leaf lesions with gray fungal sporulation; and ghost spots or white halos on fruits. Microscopically, the pathogen exhibits septate, branched mycelium (hyaline when young, darkening with age) and branched conidiophores bearing continuous or one-septate, elongated, dark conidia at their tips.
- **Prevention and Cultural Practices**: Provide adequate ventilation around plants by pruning foliage, avoid close plant spacing, and apply protective fungicides to pruning wounds.
- **Biological and Chemical Control**:
  - **Biological Control**: Not specified in the provided sources.
  - **Chemical Control**: Regular protective spraying with Bordeaux mixture (1.0%) or Mancozeb (0.2%). Applying fungicide directly to pruning wounds helps reduce infection.
  - **Resistant Varieties**: Cultivate resistant varieties such as Vetomold, or Eurocross varieties including Antincold, LMRI, and Sapsford's No. 1.
- **Treatment Methods, Pesticide Names, Doses, and Application Methods**: Spray Bordeaux mixture at 1.0% concentration or Mancozeb at 0.2% concentration as regular foliar sprays. Apply fungicide directly to fresh pruning wounds.
- **Integrated Disease Management**: Combine the use of resistant cultivars (Vetomold, Antincold, LMRI, Sapsford's No. 1), pruning and proper spacing to maximize air movement, application of fungicides to pruning wounds, and regular preventive foliar sprays of Bordeaux mixture (1.0%) or Mancozeb (0.2%).
- **Monitoring and Early Detection**: Not specified in the provided sources.
- **Effects on Crop Growth, Fruit Quality, and Yield**: Girdles and kills stems and shoots, destroying whole plants. Causes soft, watery decay of fruits, rendering them unmarketable. Ghost spots cause cosmetic disfigurement that lowers fruit market value.
- **Other Relevant Information**: Distributed worldwide. Exposure of infected fruit to direct sunlight and high temperatures arrests disease development, leaving ghost spots. No conflicting recommendations were identified in the provided sources.

1.16 Southern Blight

- **Disease Name and Causal Organism**: Southern Blight, caused by the soil-borne fungus *Sclerotium rolfsii* (teleomorph: *Athelia rolfsii*).
- **Symptoms and Signs**: Causes seedling damping-off, crown rot, root rot, and fruit rot. On seedlings, dark brown lesions form at or below the soil line, completely invading stem tissues and causing rapid damping-off and death. On older plants, lesions girdle the stem near the soil line, causing sudden wilting of the plant without an initial change in foliage color; severely infected plants eventually die. The lesion extends several centimeters above the soil line and causes root rot below the soil line. Under adequate moisture, a whitish fungal growth (mycelium) covers the lesion surface, readily producing small, tan, spherical sclerotia measuring 1–2 mm (0.06 in.) in diameter. Fruits coming into contact with infected soil develop sunken, yellowish lesions with a ruptured epidermis, covered with white mycelium and sclerotia.
- **Disease Development and Spread**: The fungus is an efficient saprophyte that persists in soil and plant debris for several years. It is spread through surface water movement and by infested soil transported on cultivation equipment.
- **Conditions Favourable for Disease Development:** High soil temperatures ranging from 30 °C to 35 °C (86 °F to 95 °F), combined with high soil moisture levels, strongly favour disease development.
- **Diagnosis and Identification:** Diagnostic features include girdling brown stem lesions at the soil line with sudden wilting of green-foliaged plants, conspicuous white fan-like mycelial growth, and small (1–2 mm) tan spherical sclerotia present on lower stems, soil surfaces, or fruit lesions.
- **Prevention and Cultural Practices**: Carefully regulate soil moisture and ensure proper drainage. Practice deep plowing of crop residues. Maintain strict field sanitation by removing and burning all infected plants. Implement a three-year crop rotation with non-host crops such as corn or sorghum.
- **Biological and Chemical Control**:
  - **Biological Control**: Not specified in the provided sources.
  - **Chemical Control**: Soil drenching with Captan 50% WP @ 1 g/litre against *Sclerotium* rot. Execute soil fumigation and apply protective fungicides in affected fields.
- **Treatment Methods, Pesticide Names, Doses, and Application Methods**: Remove and burn infected plants. Drench the soil around affected areas with Captan 50% WP at a dose of 1 g per litre of water. Perform soil fumigation and fungicide treatments as recommended for infested soils.
- **Integrated Disease Management**: Combine cultural moisture management, deep plowing of residue, prompt removal and burning of diseased plants, a 3-year crop rotation with corn or sorghum, soil drenching with Captan 50% WP (1 g/L), and soil fumigation or fungicide applications.

- **Monitoring and Early Detection**: Not specified in the provided sources.
- **Effects on Crop Growth, Fruit Quality, and Yield**: Causes rapid seedling mortality, girdles mature stems leading to complete plant collapse and death, and decays soil-contacting fruits, causing total crop loss in affected areas.
- **Other Relevant Information**: Distributed worldwide. No conflicting recommendations were identified in the provided sources.

1.17 Powdery Mildew

Disease Name and Causal Organism:

- Disease Name: Powdery Mildew
- Causal Organism: *Oidium* sp., *Oidiopsis* sp., *Leveillula taurica* (anamorph: *Oidiopsis sicula*), and *Oidium neolycopersicum*.

Symptoms and Signs:

- Initial foliar symptoms appear as light green to bright yellow chlorotic lesions on upper leaf surfaces.
- Whitish powdery fungal growth and masses of conidia form on leaf surfaces (on the lower leaf surface for *Leveillula*, and primarily on the upper leaf surface or both surfaces for *Oidium*).
- Affected leaves become dwarfed, stiff, narrow, chlorotic, and eventually necrotic, leading to leaf collapse and complete defoliation.
- Terminal growth of affected shoots is stunted or killed.

Disease Development and Spread:

- The pathogens have wide host ranges and produce abundant fungal conidia that travel long distances on air currents.
- The disease occurs severely during dry seasons in field production and spreads rapidly in greenhouse environments.

Conditions Favourable for Disease Development:

- For *Leveillula taurica*: Warm temperatures around 27°C (80°F) (conidia germinate between 10–32°C) and low relative humidity (52–75%).
- For *Oidium neolycopersicum*: Temperatures between 20–27°C (68–81°F), low light intensity, and high relative humidity (85–95%), though infection can occur at relative humidity as low as 50%.

Diagnosis and Identification:

- Identified by visual observation of white powdery fungal sporulation on leaf surfaces paired with chlorotic yellow foliar lesions.

Prevention and Cultural Practices:

- Cultivate resistant tomato varieties or cultivars.
- Adopt early planting schedules.
- Maintain proper plant spacing of approximately 3 feet apart to allow air circulation and fast drying of foliage.
- Apply organic mulch around plant bases to protect roots and prevent fungal spores from splashing onto foliage.
- Implement sprinkler irrigation and perform periodic clipping and removal of lower leaves.
- Immediately remove and destroy infected plant parts to prevent disease spread.

Biological and Chemical Control:

- Biological Control: Not specified in the provided sources.
- Chemical Control: Timely foliar applications of sulfur or Karathane fungicides with thorough foliage coverage.

Treatment Methods, Pesticide Names, Doses, and Application Methods:

- Wettable Sulphur: Applied as a foliar spray at a dose of 3 g per liter of water.
- Karathane: Applied as a foliar spray at a concentration of 0.1% twice at an interval of 10 days.

Integrated Disease Management:

- Integration of resistant varieties, wider plant spacing, organic mulching, prompt sanitation, and preventative sulfur or Karathane sprays.

Monitoring and Early Detection:

- Inspect leaf surfaces regularly for early chlorotic spots and initial white powdery fungal patches.

Effects on Crop Growth, Fruit Quality, and Yield:

- Causes stunting or death of terminal shoot growth, extensive defoliation, smaller fruit size, fruit sunscald due to loss of canopy protection, and significant reduction in fruit yield (though fruit tissue itself remains uninfected by *Oidium*).

Other Relevant Disease-Specific Information:

- Mainly represents a major production threat in greenhouse cultivation where severe defoliation leads to substantial yield losses.

1.18 Phoma Rot

Disease Name and Causal Organism:

- Disease Name: Phoma Rot
- Causal Organism: *Phoma destructiva*

Symptoms and Signs:

- Presence of small, black, pimple-like eruptions (pycnidia) on affected fruit and stem surfaces.
- Foliage develops small dark brown to black spots with concentric rings, starting on older leaves and causing severe defoliation.
- Stems develop dark brown necrotic lesions with concentric rings.
- Fruit lesions initiate at the calyx end as small sunken spots that enlarge into sunken, black, leathery lesions containing numerous black pycnidia in the center.

Disease Development and Spread:

- The pathogen is primarily seed-borne and survives in soil, infected plant debris, and alternate hosts such as pepper and closely related nightshade weeds.
- Conidia exuded from pycnidia spread via wind, rain splashes, overhead irrigation, and on workers' hands, clothing, and tools.
- Infection is facilitated by plant injuries such as pruning wounds, insect feeding, mechanical damage, or fruit cracking.

Conditions Favourable for Disease Development:

- Moderate temperatures around 20°C (68°F) and high relative humidity.
- Low soil nitrogen and phosphorus levels increase crop susceptibility.

Diagnosis and Identification:

- Identified by dark lesions with concentric rings on leaves, stems, and fruits containing distinct, minute black fungal fruiting bodies (pycnidia).

Prevention and Cultural Practices:

- Practice field sanitation by removing and destroying infected plant debris and related nightshade weeds.
- Utilize long crop rotations with non-host species.
- Maintain proper soil fertility, avoiding deficiencies in nitrogen and phosphorus.
- Avoid fruit injury during harvest and pick dry fruit to prevent pathogen spread in packing containers.
- Avoid overhead irrigation.

Biological and Chemical Control:

- Biological Control: Not specified in the provided sources.
- Chemical Control: Seed treatment with organomercurial fungicides and foliar spraying with Zineb.

Treatment Methods, Pesticide Names, Doses, and Application Methods:

- Organomercurial Fungicides: Used for seed treatment prior to planting.
- Zineb: Applied as a foliar spray at a concentration of 0.2%.

Integrated Disease Management:

- Combines disease-free seed treatment, weed eradication, long crop rotation, soil fertility management, careful harvesting practices, and protective Zineb sprays.

Monitoring and Early Detection:

- Monitor leaves and calyx ends of developing green and ripe fruits for dark spot initiation and pycnidia formation.

Effects on Crop Growth, Fruit Quality, and Yield:

- Causes severe foliar defoliation, extensive fruit decay in both green and ripe tomatoes, renders produce unmarketable, and results in heavy yield losses.

Other Relevant Disease-Specific Information:

- Ascospores are irregularly arranged in two series, ellipsoid with obtuse ends, hyaline, and guttulate; pycnidia are dark brown and can be solitary or gregarious.

1.19 Tomato Mosaic Virus (ToMV)

Disease Name and Causal Organism:

- Disease Name: Tomato Mosaic Disease
- Causal Organism: Tomato Mosaic Virus (ToMV)

Symptoms and Signs:

- Light and dark green chlorotic mottling (mosaic pattern) on foliage.
- Leaflets are distorted, puckered, reduced in size, and may show "fern leaf" symptoms (indented and curled blades).
- Initial infection causes wilting of young leaves on sunny days.
- Infected plants appear pale green, spindly, and stunted.
- Necrotic streaks or spots appear on stems, petioles, foliage, and fruit.
- Fruit shows uneven ripening and internal wall browning ("brown wall").

Disease Development and Spread:

- Mechanically transmitted via direct contact with clothes, laborers' hands, plant debris, tools, machinery, and touching between infected and healthy plants.
- Seed-borne (up to 94% of seeds can harbor the virus); infection frequently occurs during transplanting or handling.
- Possesses a wide host range including solanaceous crops and weeds; infested crop debris serves as an inoculum source.

Conditions Favourable for Disease Development:

- Cool temperatures favor the expression of "fern leaf" distortion, whereas high temperatures can mask foliar mosaic symptoms.

Diagnosis and Identification:

- Diagnosed by green foliar mottling, fern-like leaf curling, pale spindly stunting, necrotic stem streaks, and internal browning of fruit walls.

Prevention and Cultural Practices:

- Select disease-free seeds from healthy plants.
- Enforce strict sanitation: workers must wash hands with soap and water before handling plants.
- Steam sterilize potting soil, containers, tools, and equipment between cropping cycles.
- Rogue and destroy infected seedlings in nursery beds and main fields; do not transplant infected seedlings.
- Rotate crops with non-solanaceous species (avoid tobacco, potato, chili, capsicum, and brinjal).
- Avoid knife pruning; snap off suckers without touching main plant stems.
- Direct seed in the field to eliminate transplanting transmission.
- Plant ToMV-resistant tomato varieties.

Biological and Chemical Control:

- Biological Control: Not specified in the provided sources.
- Chemical Control: Chemical seed treatments are used to eliminate virus particles from seed coats.

Treatment Methods, Pesticide Names, Doses, and Application Methods:

- Trisodium Phosphate: Soak seeds in a solution of 90 g per liter of water for one day prior to sowing, followed by thorough rinsing in water and shade drying.

Integrated Disease Management:

- Integrates resistant varieties, trisodium phosphate seed soaking, strict tool/hand hygiene, non-solanaceous crop rotations, and sucker snapping.

Monitoring and Early Detection:

- Inspect nursery beds and young field plantings regularly on sunny days for leaf mottling, puckering, or temporary leaf wilting.

Effects on Crop Growth, Fruit Quality, and Yield:

- Causes severe plant stunting, pale spindly growth, internal fruit wall browning, uneven ripening, and substantial reductions in total fruit yield.

Other Relevant Disease-Specific Information:

- Virus particles are rod-shaped, straight, and non-enveloped, with a high thermal inactivation point between 85–90°C.

1.20 Leaf Mold

Disease Name and Causal Organism:

- Disease Name: Leaf Mold
- Causal Organism: *Fulvia fulva* (synonym: *Cladosporium fulvum*)

Symptoms and Signs:

- Initial symptoms show pale green to yellowish areas on the upper surface of older leaves.
- Corresponding lower leaf surfaces develop dense masses of olive-green velvety fungal sporulation.
- Leaves turn yellow, wilt, and drop prematurely (severe defoliation).
- Stems, blossoms, and fruits can become infected; infected fruit displays a black, leathery rot at the calyx end.

Disease Development and Spread:

- Pathogen survives as conidia or sclerotia in soil and plant debris for at least one year.
- Conidia are readily dispersed by wind, rain, and workers' clothing or equipment.
- Represents a major problem in greenhouse and protected cultivation systems where it spreads rapidly under favorable conditions.

Conditions Favourable for Disease Development:

- High relative humidity (90%) and warm temperatures around 24°C (75°F) are optimal; disease can occur between 10–32°C (50–90°F).
- Disease will NOT develop if relative humidity is kept below 85%.

Diagnosis and Identification:

- Identified by upper leaf yellow spots paired with olive-green velvety fungal growth on lower leaf surfaces.

Prevention and Cultural Practices:

- Provide adequate air movement, ventilation, and heating in greenhouses to maintain relative humidity strictly below 85%.
- Plant resistant tomato varieties where available.
- Practice good sanitation and prevent worker movement from infected to healthy areas to minimize spore dissemination.

Biological and Chemical Control:

- Biological Control: Not specified in the provided sources.
- Chemical Control: Application of a protective fungicide spray program.

Treatment Methods, Pesticide Names, Doses, and Application Methods:

- Fungicide Spray Program: Application of protective fungicides (specific chemical names, doses, and application methods are not specified in the provided sources).

Integrated Disease Management:

- Combines strict greenhouse humidity regulation (<85% RH), resistant varieties, sanitation practices, and protective fungicide applications.

Monitoring and Early Detection:

- Monitor the underside of older foliage in protected structures regularly for olive-green fungal sporulation.

Effects on Crop Growth, Fruit Quality, and Yield:

- Causes progressive foliar yellowing, severe defoliation, blossom abortion, calyx-end black leathery fruit rot, and major yield loss in greenhouse production.

Other Relevant Disease-Specific Information:

- Numerous physiological races of the fungus exist, making breeding and maintaining resistant varieties difficult.

1.21 Fusarium Crown and Root Rot

Disease Name and Causal Organism:

- Disease Name: Fusarium Crown and Root Rot
- Causal Organism: *Fusarium oxysporum* f. sp. *radicis-lycopersici*

Symptoms and Signs:

- Symptoms first appear at the mature green fruit stage as yellowing of the oldest leaves, progressing upward to younger foliage.
- Diurnal wilting occurs on hot sunny days; severe infections cause plants to quickly wilt and die.
- Entire root system turns brown and the taproot rots away.
- Chocolate brown lesions form at or near the soil line and extend into the vascular system.
- Vascular discoloration is brown and typically does not extend more than 25 cm (10 inches) above the soil line.
- Fungal sporulation may be observed on exposed crown lesions under high humidity.

Disease Development and Spread:

- Survives in soil, infected plant debris, and roots of alternative host crops (eggplant, pepper, and legumes) for several years.
- Fungal conidia spread via soil movement, farm machinery, irrigation water, and infected transplants.
- Infection occurs through feeder roots and wounds caused by secondary root formation.

Conditions Favourable for Disease Development:

- Favored by cool soil temperatures around 20°C (68°F).

Diagnosis and Identification:

- Identified by mature-stage leaf yellowing and wilting, taproot decay, chocolate brown crown lesions, and vascular discoloration restricted to within 25 cm of the soil line (distinguishes it from Fusarium wilt).

Prevention and Cultural Practices:

- Utilize resistant tomato varieties (most effective control measure).
- Practice thorough sanitation of farm machinery, tools, and irrigation water.
- Avoid wounding roots during cultivation and field operations.

Biological and Chemical Control:

- Biological Control: Not specified in the provided sources.
- Chemical Control: Soil steaming combined with fungicide drenches. Chemical soil fumigation is not effective.

Treatment Methods, Pesticide Names, Doses, and Application Methods:

- Soil Steaming and Fungicide Drenching: Combined soil steaming and fungicide drenches reduce losses (specific fungicide names and doses are not specified in the provided sources).

- Soil Fumigation: Not recommended as an effective control because *Fusarium oxysporum* f. sp. *radicis-lycopersici* rapidly recolonizes sterilized soil.

Integrated Disease Management:

- Integrates resistant cultivars, soil steaming, machinery sanitation, root health maintenance, and proper irrigation management.

Monitoring and Early Detection:

- Inspect plants at mature green fruit stage for midday wilting and lower leaf chlorosis, followed by checking the stem base for brown crown lesions.

Effects on Crop Growth, Fruit Quality, and Yield:

- Leads to sudden wilting and premature plant death during fruit bearing, destruction of root systems, and total loss of fruit production on affected plants.

Other Relevant Disease-Specific Information:

- The fungus rapidly recolonizes fumigated or pasteurized soil, making standard soil fumigation ineffective for disease control.

1.22 Gray Leaf Spot

- **Disease Name and Causal Organism:** Gray Leaf Spot, caused by the fungal pathogens *Stemphylium solani* (synonym: *Stemphylium floridanum*) and *Stemphylium botryosum* f. sp. *lycopersici*.
- **Symptoms and Signs:** Foliar lesions begin as small, brownish-black specks that enlarge into grayish-brown, glazed angular spots approximately 3 mm in diameter, frequently surrounded by a chlorotic yellow halo. Lesions eventually dry out and develop central cracks. Extensive lesion development leads to leaf yellowing, severe leaf drop, and defoliation. Fruits and stems are not affected by this pathogen.
- **Disease Development and Spread:** The fungus overwinters and survives in soil, infected plant debris, volunteer tomato plants, alternate solanaceous crops, and weeds. Infected transplants act as major inoculum sources. Fungal spores are disseminated across fields by wind and rain splashing.
- **Conditions Favourable for Disease Development:** Warm, humid, or wet weather, prolonged leaf wetness or dew periods, and overhead sprinkler irrigation create optimal conditions for infection.
- **Diagnosis and Identification:** Diagnosed by the presence of grayish-brown, glazed, angular leaf spots with yellow halos and center cracking, accompanied by progressive foliar defoliation while fruit and stem tissues remain completely symptom-free.
- **Prevention and Cultural Practices:** The primary control measure is the widespread cultivation of resistant tomato varieties. Managing solanaceous weeds and avoiding overhead sprinkler irrigation also reduce disease establishment.

- **Biological and Chemical Control:** Protective fungicide application programs are necessary when susceptible tomato cultivars are grown.
- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Specific fungicide product names, exact chemical doses, and spray application schedules are not specified in the provided sources.
- **Integrated Disease Management:** Combining resistant plant cultivars with field sanitation, crop rotation, weed management, and targeted protective fungicide sprays.
- **Monitoring and Early Detection:** Regularly inspecting lower foliage for brownish-black specks during warm, humid weather or following overhead irrigation.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes significant leaf chlorosis, premature leaf drop, and extensive defoliation, reducing overall plant vigor, though fruits and stems sustain no direct damage.
- **Other Relevant Information:** Widespread commercial adoption of resistant tomato varieties has significantly minimized the economic impact of this disease.

1.23 Anthracnose

- **Disease Name and Causal Organism:** Anthracnose, caused by fungal pathogens including *Colletotrichum coccodes*, *Colletotrichum dematium*, *Colletotrichum gloeosporioides*, and *Colletotrichum phomoides* (teleomorph: *Glomerella cingulata*).
- **Symptoms and Signs:** On green and ripening fruit, initial symptoms appear as small, circular, slightly sunken, water-soaked spots. Lesions expand, become depressed, and exhibit distinct dark concentric rings. In humid weather, masses of gelatinous pink conidia accumulate on lesion surfaces alongside dark, pimple-like microsclerotia. Infection occurs on green fruit but remains latent until ripening. Root infections manifest as brown lesions with microsclerotia (black dot root rot). Leaf symptoms are uncommon and appear as small circular brown spots surrounded by yellow halos.
- **Disease Development and Spread:** The fungus persists in soil and decaying crop residue for several years as microsclerotia, possessing a broad host range of over 68 species. Fungal spores infect host tissues directly or through mechanical wounds, spreading via rain splashing and overhead irrigation.
- **Conditions Favourable for Disease Development:** Temperatures between 10–30°C (50–86°F), presence of free moisture, splashing rain, overhead irrigation, and plants experiencing nutritional stress or root damage.
- **Diagnosis and Identification:** Identified by depressed circular fruit lesions with dark concentric rings, slimy pink spore masses, and microsclerotia on fruit or root surfaces.
- **Prevention and Cultural Practices:** Selecting well-drained planting sites, practicing crop rotation with non-host plant species, applying broad-spectrum soil fumigants, and preventing mechanical root injury.

- **Biological and Chemical Control:** Initiating a preventive fungicide spray program at the first green fruit stage and continuing applications through harvest.
- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Specific fungicide product names, chemical doses, and application methods are not specified in the provided sources.
- **Integrated Disease Management:** Integrating proper field drainage, crop rotation, soil management, avoidance of fruit and root wounding, and protective fungicide schedules.
- **Monitoring and Early Detection:** Examining developing green and ripening fruits for small water-soaked depressions, particularly on lower fruit clusters.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes severe fruit decay, internal fruit tissue breakdown, root rot, and severe post-harvest economic loss.
- **Other Relevant Information:** Conflicting nomenclature exists across sources regarding the primary causal species (*Colletotrichum coccodes* versus *Colletotrichum phomoides*).

1.24 Bacterial Speck

- **Disease Name and Causal Organism:** Bacterial Speck, caused by the bacterium *Pseudomonas syringae* pv. *tomato* (races 0 and 1).
- **Symptoms and Signs:** Attacks leaves, stems, petioles, and flower parts. Leaves exhibit dark brown to black spots surrounded by chlorotic yellow halos, which coalesce to kill broad areas of leaf tissue. Dark lesions with yellow margins form on leaf edges where guttation fluid collects. Stems and petioles develop oval to elongated black lesions. Green fruits show tiny (1 mm), dark, superficial, speck-like lesions surrounded by green halos.
- **Disease Development and Spread:** Favored by cool weather and high moisture, requiring approximately one day of leaf wetness to establish infection. The bacterium survives on plant roots, foliage of non-host crops, weeds, and occasionally on seeds.
- **Conditions Favourable for Disease Development:** Cool temperatures between 13–25°C (55–77°F), rainy weather, heavy dew, and overhead sprinkler irrigation.
- **Diagnosis and Identification:** Confirmed by small, superficial black specks with yellow halos on foliage and green-haloed superficial specks on immature green fruit.
- **Prevention and Cultural Practices:** Planting resistant tomato cultivars is the most effective defense. Utilizing furrow or drip irrigation instead of overhead sprinkler irrigation significantly reduces disease occurrence.
- **Biological and Chemical Control:** Applying protective copper sprays early in the growing cycle helps suppress disease development.

- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Specific copper formulations, exact doses, and application schedules are not specified in the provided sources.
- **Integrated Disease Management:** Combining genetic host resistance, drip/furrow irrigation management, weed control, and preventive early-season copper sprays.
- **Monitoring and Early Detection:** Inspecting leaf surfaces, stem petioles, and young green fruit for chlorotic-haloed specks during cool, wet spring weather.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Induces severe leaf necrosis and defoliation, blighting of flowers, and cosmetic fruit blemishes that reduce fresh-market value.
- **Other Relevant Information:** The disease rarely develops or causes economic damage in arid cultivation areas using drip or furrow irrigation systems.

1.25 Syringae Leaf Spot

- **Disease Name and Causal Organism:** Syringae Leaf Spot, caused by the bacterium *Pseudomonas syringae* pv. *syringae*.
- **Symptoms and Signs:** Foliar symptoms vary from brown leaf spots lacking halos to dark brown or black spots with bright yellow chlorotic halos, visually resembling bacterial speck. Leaf spots can grow larger than typical bacterial speck lesions.
- **Disease Development and Spread:** The causal bacterium is a weak pathogen that persists on host and non-host foliage in a non-parasitic state. It spreads during cool, wet weather and requires mechanical wounding or prior disease lesions to infect plant tissues.
- **Conditions Favourable for Disease Development:** Cool, wet environmental conditions, high moisture, and mechanical or wind-driven plant wounding.
- **Diagnosis and Identification:** Requires laboratory isolation and testing to accurately differentiate the pathogen from *Pseudomonas syringae* pv. *tomato* due to overlapping visual symptoms.
- **Prevention and Cultural Practices:** Minimizing mechanical wounding, sandblasting, and equipment damage to plants during field management operations.
- **Biological and Chemical Control:** Chemical control can be achieved using copper sprays, though spraying is usually unnecessary because economic losses are rare.
- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Specific bactericide product names, chemical doses, and spray application schedules are not specified in the provided sources.
- **Integrated Disease Management:** Utilizing laboratory diagnostic confirmation prior to applying strict chemical controls to prevent unnecessary pesticide sprays.

- **Monitoring and Early Detection:** Monitoring fields for leaf spot expansion following cold rainstorms or mechanical damage events.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes minor foliar spotting with minimal overall impact on plant growth, fruit quality, or crop yield.
- **Other Relevant Information:** Spraying is generally omitted because the pathogen is a weak invader that seldom causes severe crop loss.

1.26 Pith Necrosis

- **Disease Name and Causal Organism:** Pith Necrosis, caused by bacterial pathogens *Pseudomonas corrugata* and *Pseudomonas mediterranea*.
- **Symptoms and Signs:** Begins as yellowing and wilting of young leaves in the upper plant canopy. Dark brown to black lesions form on exterior stem surfaces. Longitudinal sections of infected stems reveal dark brown discoloration of the pith and vascular tissues, along with hollow or segmented cavities. Adventitious roots develop profusely along affected stem sections.
- **Disease Development and Spread:** Symptom expression typically coincides with fruit growth reaching the mature green stage, particularly in lush, overly succulent plants. Bacteria spread through pruning, field handling, and mechanical wounding when foliage is wet.
- **Conditions Favourable for Disease Development:** Cool night temperatures, high humidity, excessive nitrogen fertilization, rank plant succulence, and handling foliage while wet.
- **Diagnosis and Identification:** Identified by upper-canopy leaf yellowing, stem exterior blackening, internal dark brown necrotic/hollowed stem pith, and extensive adventitious root formation.
- **Prevention and Cultural Practices:** Avoiding excessive nitrogen fertilizer applications, refraining from pruning or handling plants when foliage is wet, and routinely disinfecting pruning tools.
- **Biological and Chemical Control:** Not specified in the provided sources.
- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Not specified in the provided sources.
- **Integrated Disease Management:** Combining balanced nitrogen fertilization, strict tool disinfection, minimal wounding, and avoiding wet-crop cultural operations.
- **Monitoring and Early Detection:** Checking upper leaves, stem bases, and internal stem pith around the onset of mature green fruit development.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes severe stem collapse, vascular breakdown, plant wilting, and complete crop death under severe infection.

- **Other Relevant Information:** Disinfecting pruning tools between plants significantly reduces the transmission rate of the bacteria across the field.

1.27 Big Bud

- **Disease Name and Causal Organism:** Big Bud, caused by a Phytoplasma pathogen transmitted by the brown leafhopper (*Orosius argentatus*).
- **Symptoms and Signs:** Characterized by abnormal enlargement and swelling of flower buds. Sepals remain fused, forming an enlarged, green, bladder-like structure with a toothed top opening, while buds point vertically. Axillary buds proliferate into small side shoots, internodes shorten, stems thicken, and aerial roots form. Leaves become reduced, yellow-green, and margins roll inward. Fruit set is severely reduced; existing green fruits become hard, tough, deformed, and fail to ripen or color normally.
- **Disease Development and Spread:** The phytoplasma overwinters and survives in alternate crops (peppers, lettuce, eggplant, potato) and weed species (dock, nightshade, sowthistle, lamb's-quarter, jimsonweed). Vector leafhoppers transmit the pathogen when migrating into tomato fields.
- **Conditions Favourable for Disease Development:** High populations and seasonal migration of viruliferous brown leafhoppers from adjacent weed or crop hosts into tomato fields.
- **Diagnosis and Identification:** Diagnosed by swollen, bladder-like green calyxes pointing vertically, extreme axillary shoot proliferation, stem thickening, and hard, tough, unripened fruit.
- **Prevention and Cultural Practices:** Prompt removal and destruction of infected plant parts, alongside eradication of host weeds adjacent to tomato fields.
- **Biological and Chemical Control:** Vector leafhopper populations can be controlled through an insecticide spray program.
- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Specific insecticide product names, dosages, and application methods are not specified in the provided sources.
- **Integrated Disease Management:** Combining host weed destruction, immediate roguing of infected crop plants, and vector leafhopper chemical control programs.
- **Monitoring and Early Detection:** Regularly examining actively growing shoot tips and developing flower trusses for upright, swollen green flower buds.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Induces severe plant stunting, abnormal shoot bushy growth, poor blossom set, and unmarketable hard, misshapen fruit.
- **Other Relevant Information:** The disease is generally considered to be of occasional economic importance in tomato production.

1.28 Tomato Bunch Top Virus

- **Disease Name and Causal Organism:** Tomato Bunch Top Virus (TBTV).
- **Symptoms and Signs:** Infected plants display extensive abnormal growth with apical proliferation. Emerging leaves from axillary buds create a closely crowded, bunchy appearance at shoot tips. Leaflet margins curl inward toward the tips, and leaf blades appear puckered. Stem and leaf necrosis is common. Plants set very few flowers and produce only one or two extremely small fruits.
- **Disease Development and Spread:** The virus persists in and spreads from alternate or collateral weed hosts present in and surrounding tomato fields.
- **Conditions Favourable for Disease Development:** Presence of unmanaged alternate or collateral weed reservoirs near tomato fields.
- **Diagnosis and Identification:** Identified by crowded, bunchy apical leaf clusters, inwardly curled leaflet margins, puckered leaf blades, stem/leaf necrosis, and extreme fruit yield loss.
- **Prevention and Cultural Practices:** Complete roguing and destruction of infected tomato plants, combined with the destruction of alternate and collateral weed hosts during weeding and earthing-up operations.
- **Biological and Chemical Control:** Not specified in the provided sources.
- **Treatment Methods, Pesticide Names, Doses, and Application Methods:** Not specified in the provided sources.
- **Integrated Disease Management:** Combining systematic roguing of infected plants with rigorous field sanitation and weed host destruction.
- **Monitoring and Early Detection:** Inspecting growing points for bunchy, puckered leaf proliferation and margin curling.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes severe plant stunting and drastic yield reduction, leaving plants almost entirely barren.
- **Other Relevant Information:** Regular weed host destruction during field cultivation operations is essential to restrict viral spread.

1.29 Black Mold Rot

- **Causal Organism:** *Alternaria alternata* and *Stemphylium* species.
- **Symptoms and Signs:** Symptoms range from superficial flecking to dark brown-black, dry, sunken lesions that may extend deeply into fruit cavities. V-shaped lesions frequently develop on the stem scar. Under humid conditions, a dense layer of black fungal conidia forms over the lesion surface.
- **Disease Development and Spread:** Fungi enter the fruit through natural openings such as stem scars, growth cracks, or wounds caused by insect feeding and mechanical handling. Infection develops when warm temperatures coincide with elevated humidity.

- **Conditions Favourable for Disease Development:** Warm temperatures and high humidity.
- **Diagnosis and Identification:** Superficial flecks progressing to brown-black dry sunken lesions, V-shaped stem scar lesions, and a dense black conidial layer under high humidity.
- **Prevention and Cultural Practices:** Avoid mechanical injury to fruits during harvesting. Practice cultural methods that prevent fruit contact with the soil. Improve field air movement by orienting rows with prevailing winds and maintaining proper row and plant spacing. Schedule sprinkler irrigation to minimize the duration of free water on fruit surfaces. Select firm-fruited or crack-resistant varieties.
- **Biological and Chemical Control:** Broad-spectrum fungicides may offer partial protection. Specific pesticide names, doses, and application methods are not specified in the provided sources.
- **Treatment Methods:** Not specified in the provided sources.
- **Integrated Disease Management:** Combine crack-resistant cultivars, proper irrigation timing, optimal field ventilation, careful harvest handling, and sanitation.
- **Monitoring and Early Detection:** Inspect ripening fruit for early superficial flecking or stem scar discoloration during warm, humid weather.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes dry sunken lesions and cavity rot in fruits, severely compromising marketability and crop value.
- **Other Relevant Information:** Susceptibility increases as fruits mature or when skin cracks develop after environmental stress.

1.30 Rhizoctonia Fruit Rot

- **Causal Organism:** *Rhizoctonia solani*.
- **Symptoms and Signs:** Begins as a firm rot that rapidly turns into a watery soft rot. It typically attacks ripe fruit in direct contact with the soil. The infected area forms distinct concentric rings, and a brown fungal growth frequently covers the surface of the lesion.
- **Disease Development and Spread:** Pathogens enter through natural openings or fruit wounds when ripe fruit comes into physical contact with infested soil.
- **Conditions Favourable for Disease Development:** Warm temperatures, high soil moisture, elevated humidity, and direct soil contact.
- **Diagnosis and Identification:** Firm rot turning watery soft on soil-touching ripe fruit, presence of concentric ring patterns, and brownish surface fungal growth.
- **Prevention and Cultural Practices:** Implement cultural practices that prevent fruit from touching the soil. Orient rows in the direction of prevailing winds, maintain proper plant spacing, and schedule sprinkler irrigation to shorten leaf and fruit wetness duration. Prevent fruit skin injury during handling.

- **Biological and Chemical Control:** Fungicide programs may provide partial protection. Specific chemical names, doses, and application methods for fruit rot control are not specified in the provided sources.
- **Treatment Methods:** Not specified in the provided sources.
- **Integrated Disease Management:** Combine fruit elevation, optimal field drainage, proper row spacing, balanced irrigation, and careful handling.
- **Monitoring and Early Detection:** Inspect low-hanging ripe fruits touching the ground for initial firm or ringed lesions.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes rapid soft watery decay of mature fruits, destroying harvested yield.
- **Other Relevant Information:** Direct contact between mature fruit and moist soil is the primary driver of infection.

1.31 Rhizopus Rot

- **Causal Organism:** *Rhizopus stolonifer*.
- **Symptoms and Signs:** Lesions develop rapidly with a puffy, water-soaked appearance. A white fungal growth interspersed with tiny black spheres on raised stalks spreads over the infected tissue. Affected fruits produce a distinct fermented odor, distinguishing it from bacterial soft rot and sour rot.
- **Disease Development and Spread:** The fungus enters through growth cracks, wounds, or natural openings under warm and humid conditions.
- **Conditions Favourable for Disease Development:** Warm temperatures and high humidity.
- **Diagnosis and Identification:** Puffy, water-soaked lesions accompanied by white fungal growth with tiny black spheres on stalks and a characteristic fermented odor.
- **Prevention and Cultural Practices:** Avoid fruit injury during harvest and packing. Prevent fruit from contacting the soil, improve field air circulation, and adjust irrigation schedules to minimize surface moisture.
- **Biological and Chemical Control:** Fungicides may offer limited protection. Specific pesticide names, doses, and application guidelines are not specified in the provided sources.
- **Treatment Methods:** Not specified in the provided sources.
- **Integrated Disease Management:** Focus on harvest sanitation, injury prevention, humidity control, and proper fruit handling.
- **Monitoring and Early Detection:** Monitor harvested and field fruits for soft puffy spots and white-and-black spore growth during warm weather.

- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes rapid fruit decay and liquefaction, rendering infected harvests entirely unmarketable.
- **Other Relevant Information:** The presence of a fermented odor rather than a putrid odor is a key diagnostic distinction.

1.32 Pythium Fruit Rot

- **Causal Organism:** *Pythium* species.
- **Symptoms and Signs:** Begins as water-soaked spots on both green and ripe fruit. Lesions expand rapidly, engulfing the entire fruit and imparting a swollen water-balloon appearance. When the epidermis ruptures, the fruit quickly collapses. High humidity promotes a white cottony fungal growth over the lesion.
- **Disease Development and Spread:** Infection occurs when fruits touch wet, infested soil or free water, allowing fungal entrance through wounds or unbroken epidermis.
- **Conditions Favourable for Disease Development:** Warm temperatures, high humidity, excessive soil moisture, and standing water.
- **Diagnosis and Identification:** Water-soaked spots expanding to a water-balloon-like fruit inflation, rapid fruit collapse upon skin rupture, and white cottony mycelium in humid conditions.
- **Prevention and Cultural Practices:** Prevent fruits from touching the ground. Orient planting rows with prevailing winds, maintain wide plant spacing, and schedule sprinkler irrigation to reduce moisture accumulation. Select firm-fruited or crack-resistant varieties.
- **Biological and Chemical Control:** Fungicide applications may offer partial protection. Specific pesticide names, doses, and application methods for fruit rot are not specified in the provided sources.
- **Treatment Methods:** Not specified in the provided sources.
- **Integrated Disease Management:** Integrate field drainage improvements, plant staking, soil mulching, controlled irrigation, and foliage management.
- **Monitoring and Early Detection:** Inspect lower green and ripe fruits for water-soaked spots following heavy rain or overhead irrigation.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes complete breakdown and destruction of infected green and mature fruits.
- **Other Relevant Information:** Unlike many other fruit rots, Pythium rapidly infects green immature fruits as well as ripe ones.

1.33 Sour Rot

- **Causal Organism:** *Geotrichum candidum.*
- **Symptoms and Signs:** Attacks mature green and ripe tomatoes. Decay usually originates at the stem scar and extends in sectors down the side of the fruit. A dull white fungal growth develops where the skin cracks. The fruit frequently remains firm until decay is advanced, at which point a strong sour odor develops.
- **Disease Development and Spread:** Infection progresses rapidly on mature fruits under warm, humid conditions, particularly when skin cracks or mechanical injuries are present.
- **Conditions Favourable for Disease Development:** Warm temperatures, high humidity, fruit cracking, and mechanical skin wounds.
- **Diagnosis and Identification:** Sectorial decay starting from the stem scar, dull white fungal growth in skin cracks, delayed fruit softening, and a strong sour odor.
- **Prevention and Cultural Practices:** Select tomato cultivars tolerant to growth cracking. Avoid mechanical skin injuries and regulate irrigation to prevent rapid fruit growth surges that cause skin splitting.
- **Biological and Chemical Control:** Fungicides may provide limited protection. Specific pesticide names, doses, and application protocols are not specified in the provided sources.
- **Treatment Methods:** Not specified in the provided sources.
- **Integrated Disease Management:** Combine crack-tolerant varieties, uniform water management, careful harvesting, and field sanitation.
- **Monitoring and Early Detection:** Check stem scars of ripening fruit for sectorial discoloration or white fungal growth following wet periods.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes severe fruit rotting and renders affected produce unmarketable due to sour decay.
- **Other Relevant Information:** Affected fruit retains structural firmness significantly longer than in bacterial or soft fungal rots.

1.34 Alfalfa Mosaic Virus

- **Causal Organism:** Alfalfa Mosaic Virus (AMV).
- **Symptoms and Signs:** Not specified in the provided sources.
- **Disease Development and Spread:** Transmitted by aphid vectors, including the green peach aphid (*Myzus persicae*) and potato aphid (*Macrosiphum euphorbiae*), in a non-persistent manner. Aphids acquire the virus within seconds while probing infected tissues and transmit it to healthy plants during subsequent feeding bites.
- **Conditions Favourable for Disease Development:** High aphid vector populations and proximity to infected weed or crop reservoirs.

- **Diagnosis and Identification:** Not specified in the provided sources.
- **Prevention and Cultural Practices:** Managing aphid vector populations and eliminating weed hosts near tomato fields.
- **Biological and Chemical Control:** Conserving biological control predators such as lady beetles, lacewings, and mirid bugs to reduce aphid numbers. Specific pesticide names, doses, and application methods are not specified in the provided sources.
- **Treatment Methods:** Not specified in the provided sources.
- **Integrated Disease Management:** Combine aphid vector monitoring, weed host clearance, and preservation of natural insect predators.
- **Monitoring and Early Detection:** Inspect leaf undersides and young growth regularly for aphid vector colonization.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Not specified in the provided sources.
- **Other Relevant Information:** Non-persistent transmission allows the virus to spread extremely rapidly across fields during aphid probing.

1.35 Chino del Tomate Virus

- **Causal Organism:** Chino del tomate virus (CdTV).
- **Symptoms and Signs:** Characterized by chino (curling) and rolling of leaves, interveinal yellowing of newly infected leaves, and purpling of older foliage. Plants infected at an early stage suffer severe stunting and distortion, leading to dramatically reduced fruit set.
- **Disease Development and Spread:** Virus possesses a broad host range including beans, peppers, tomatoes, and various malvaceous and solanaceous weeds. Transmitted persistently by whiteflies (*Bemisia tabaci*). Peak transmission occurs when whitefly vector populations are high, resulting in up to 100% crop infection. Mechanical transmission is not demonstrated.
- **Conditions Favourable for Disease Development:** Elevated whitefly vector populations and presence of nearby infected weed or crop hosts.
- **Diagnosis and Identification:** Foliar curling and rolling, interveinal chlorosis on young leaves, leaf purpling on mature foliage, and severe plant stunting.
- **Prevention and Cultural Practices:** Establish crop-free periods in production regions, rogue and destroy infected plants promptly, and eliminate surrounding weed host reservoirs.
- **Biological and Chemical Control:** Apply systemic insecticides to control whitefly populations. Non-systemic insecticides are generally ineffective due to difficulty in achieving coverage on lower leaf surfaces. Specific active ingredients and doses are not specified in the provided sources.

- **Treatment Methods:** Systemic insecticide treatments targeted at vector management. Specific pesticide names and application doses are not specified in the provided sources.
- **Integrated Disease Management:** Combine crop-free periods, rigorous roguing, weed host elimination, and rotational systemic insecticide applications to manage vector populations and prevent chemical resistance.
- **Monitoring and Early Detection:** Monitor whitefly vector activity on leaf undersides and inspect new foliage for interveinal yellowing and curling symptoms.
- **Effects on Crop Growth, Fruit Quality, and Yield:** Causes severe plant stunting, leaf distortion, and dramatic loss of fruit set and overall marketable yield.
- **Other Relevant Information:** Vector control is critical because whiteflies transmit the virus persistently throughout their adult lifespan.

1.36 Cucumber Mosaic Virus

Disease Name and Causal Organism:

- Disease Name: Cucumber Mosaic Virus (Cucumber Mosaic).
- Causal Organism: Cucumber mosaic virus (CMV); several strains have been reported.

Symptoms and Signs:

- Symptoms vary greatly depending on the virus strain involved.
- Causes plant stunting, often imparting a bushy appearance.
- Leaf symptoms range from mild green mottling to chlorosis or severe necrosis.
- Severe "shoestring" symptom occurs where leaf blades are greatly suppressed, leaving mostly the central rib of the leaflet.
- Fruits are reduced in size and often misshapen.

Disease Development and Spread:

- Transmitted in a non-persistent manner by aphid vectors, including *Aphis gossypii* and *Myzus persicae*.
- Virus is acquired from a wide host range comprising over 800 plant species.
- Can be mechanically transmitted; however, because it is an unstable virus, transmission by greenhouse workers and tools is much lower than with Tomato mosaic virus.
- Spreads readily from plant to plant by aphids once introduced into greenhouses or where infected alternative hosts survive year-round.

Conditions Favourable for Disease Development:

- Survival of infected alternative weed or ornamental hosts year-round near tomato crops.
- High populations and active movement of aphid vectors, particularly in greenhouse environments.

Diagnosis and Identification:

- Visual identification is based on shoestringing leaves, bushy stunting, and reduced fruit size; shoestring symptoms differ from Tomato mosaic virus by showing greater leaf blade suppression.
- Diagnostic laboratory testing procedures are not specified in the provided sources.

Prevention and Cultural Practices:

- Elimination of weeds and ornamental plants harboring the virus near fields and greenhouses.
- Roguing (removing and destroying) infected tomato plants to reduce virus spread and disease incidence.

Biological and Chemical Control:

- Controlling aphid populations in greenhouses greatly helps reduce disease incidence.
- Specific chemical pesticide names, doses, application methods, or biological control agents are not specified in the provided sources.

Integrated Disease Management and Monitoring:

- Combining weed elimination, roguing of diseased plants, and aphid vector control.
- Regular monitoring of aphid populations and routine inspection of foliage for early shoestringing and mottling symptoms.

Effects on Crop Growth, Fruit Quality, and Yield:

- Restricts plant growth, causing bushy stunting.
- Reduces fruit size and creates misshapen fruits, negatively impacting crop yield and marketability.

Other Relevant Information:

- Geographic distribution is worldwide.

1.37 Curly Top Virus

Disease Name and Causal Organism:

- Disease Name: Curly Top Virus (Curly Top).
- Causal Organism: Curly top virus (CTV), also known as beet curly top virus (BCTV); many strains have been reported.

Symptoms and Signs:

- Infected plants exhibit an erect and stunted appearance; severely infected seedlings may die.
- Leaves thicken with margins rolling upward while petioles curve downward.
- Leaves later turn dull yellow accompanied by purpling of the leaf veins.
- Very few fruits are produced; fruits set prior to infection ripen prematurely and appear dull, small, wrinkled, and dried out.

Disease Development and Spread:

- Transmitted in a persistent manner exclusively by beet leafhoppers (*Circulifer tenellus* and *C. opacipennis*).
- Possesses a wide host range of over 300 species, including sugar beet, watermelon, cantaloupe, squash, pepper, spinach, and beans.
- Virus-carrying leafhoppers move via wind into tomato fields or migrate from overwintering weedy hosts in spring.
- In-field infection patterns reflect a "raining" of virus-carrying leafhoppers; little to no secondary spread occurs within a tomato field.

Conditions Favourable for Disease Development:

- Proximity to sugar beet fields, range land, or overwintering weed hosts carrying leafhopper vectors.
- Warm spring conditions favoring wind-driven migration of beet leafhoppers into tomato fields.

Diagnosis and Identification:

- Visual diagnosis based on erect stunted growth, upward leaf rolling, petiole downward curving, vein purpling, and dull wrinkled fruit.
- Laboratory diagnostic techniques are not specified in the provided sources.

Prevention and Cultural Practices:

- Avoiding planting tomatoes near sugar beet fields and range land.
- Implementing double-row planting of processing tomatoes in affected areas (discourages leafhopper visits or allows plants to escape infection).

Biological and Chemical Control:

- Spraying insecticides on weeds outside the field to control leafhopper populations.
- Spraying tomato fields directly for leafhoppers generally will not control curly top.
- Specific chemical pesticide names, doses, or biological control agents are not specified in the provided sources.

Integrated Disease Management and Monitoring:

- Monitoring leafhopper populations in surrounding weed hosts and range lands.
- Combining spatial isolation from host crops, weed vector control, and double-row planting strategies.

Effects on Crop Growth, Fruit Quality, and Yield:

- Causes severe plant stunting and seedling mortality.
- Drastically reduces fruit set; fruits ripen prematurely and remain small, dull, wrinkled, and unmarketable.

Other Relevant Information:

- Geographic distribution includes Canada, the Mediterranean Region, Mexico, and the USA.

1.38 Tobacco Etch Virus

Disease Name and Causal Organism:

- Disease Name: Tobacco Etch Virus (Tobacco Etch).
- Causal Organism: Tobacco etch virus (TEV).

Symptoms and Signs:

- Symptoms range from mild mottling and slight foliage distortion to severe mottling and leaf crinkling.
- Fruits are frequently reduced in size, mottled, and misshapen.
- Plant stunting becomes more severe the earlier infection occurs in development.

Disease Development and Spread:

- Transmitted in a non-persistent manner by at least 10 species of aphids, including *Myzus persicae*.
- Virus-carrying aphids move into tomato fields from nearby solanaceous weeds, pepper fields, or infected tomato fields.
- Secondary spread occurs through aphid vectors or mechanically during handling, staking, and pruning of infected plants.

- Many weed species act as natural hosts for the virus.

Conditions Favourable for Disease Development:

- Proximity to infected pepper crops, solanaceous weeds, or nearby infected tomato fields.
- Active flight and movement of aphid vectors during the growing season.

Diagnosis and Identification:

- Visual identification based on foliage mottling, leaf crinkling, and fruit mottle and distortion.
- Specific diagnostic assay protocols are not specified in the provided sources.

Prevention and Cultural Practices:

- Removing weed hosts that harbor the virus in and around fields.
- Avoiding locating tomato production near pepper crops, which serve as a major virus reservoir.
- Practicing proper sanitation during staking, pruning, and plant handling to minimize mechanical spread.

Biological and Chemical Control:

- Reducing disease spread by controlling aphid vector populations is very difficult and generally not practical.
- Specific chemical pesticide names, doses, or biological controls are not specified in the provided sources.

Integrated Disease Management and Monitoring:

- Eradication of reservoir weeds combined with crop isolation from pepper fields and clean cultural practices.
- Regular visual monitoring for early foliage distortion and aphid activity.

Effects on Crop Growth, Fruit Quality, and Yield:

- Causes significant plant stunting and yield reduction, with earlier infections causing greater damage.
- Produces small, mottled, misshapen fruits that impair crop quality.

Other Relevant Information:

- Geographic distribution includes North America and South America.

1.39 Tomato Bushy Stunt Virus

Disease Name and Causal Organism:

- Disease Name: Tomato Bushy Stunt Virus (Tomato Bushy Stunt).
- Causal Organism: Tomato bushy stunt virus (TBSV).

Symptoms and Signs:

- Initial leaf symptoms vary from irregular white banding patterns to necrotic lesions or rings.
- Leaves turn from green to pale yellow with interspersed green areas; affected leaves may drop.
- Newly emerging leaves are frequently twisted with necrotic tips.
- Abundance of side shoots develops, giving the plant a bushy, stunted appearance.
- Lower leaves become chlorotic and purple as the disease progresses.
- Excessive fertilizer on young plants can cause soft stems that develop lesions at the soil line, girdling and killing the seedling.
- Fruit symptoms range from chlorotic blotches to rings or line patterns.

Disease Development and Spread:

- Mechanically transmitted; natural biological vectors are currently unknown.
- Soil-borne virus that is readily transmissible in water.
- Infects tomato plants through wounds in the roots.
- Has been detected in river water, indicating potential spread through irrigation water.

Conditions Favourable for Disease Development:

- Presence of the virus in soil or contaminated irrigation water.
- Root injuries or mechanical wounding during transplanting and cultivation.
- High nitrogen fertilizer applications on young plants leading to soft stem tissue.

Diagnosis and Identification:

- Visual identification based on white leaf banding, chlorotic or purple lower leaves, twisted tip leaves, bushy growth habit, and stem girdling lesions.
- Specific diagnostic testing methods are not specified in the provided sources.

Prevention and Cultural Practices:

- Avoiding planting tomato crops in soils known to contain the virus.

- Avoiding root injury and preventing the use of contaminated irrigation water sources.

Biological and Chemical Control:

- Once present in the soil, this virus is difficult to control.
- Specific chemical pesticide names, doses, or biological control agents are not specified in the provided sources.

Integrated Disease Management and Monitoring:

- Soil testing and field history evaluation prior to planting.
- Monitoring young plants for stem lesions at the soil line and foliage twisting or banding.

Effects on Crop Growth, Fruit Quality, and Yield:

- Causes severe bushy stunting, leaf loss, and seedling death from stem girdling.
- Reduces fruit quality and yield due to chlorotic blotches, rings, and line markings on fruits.

Other Relevant Information:

- Geographic distribution includes Argentina, Britain, California, Mexico, Morocco, and Tunisia.
- TBSV is a very stable virus with a diverse natural host range.

1.40 Tomato Double Virus Streak

Disease Name and Causal Organism:

- Disease Name: Tomato Double Virus Streak (Double Streak).
- Causal Organism: Co-infection of Tomato mosaic virus (ToMV) combined with Potato virus X (PVX).

Symptoms and Signs:

- Small brown spots develop on young leaves.
- Petioles and stems show narrow, dark brown streaks.
- Leaf lesions coalesce to form large dead (necrotic) areas, causing leaves to curl downward.
- Fruit lesions are small and skin-deep, but may run together forming large lesions with a greasy appearance, eventually becoming rough and misshapen.

Disease Development and Spread:

- Mechanically transmitted by workers, tools, and cultivation equipment.

- Occurs only when both ToMV and PVX are present in the same plant (e.g., when young plants already infected with ToMV become infected with PVX).
- Disease severity is influenced by virus strains, plant age, and day length.

Conditions Favourable for Disease Development:

- Simultaneous presence of ToMV and PVX in the crop or field environment.
- Handling potato plants prior to working with tomatoes, or planting in fields previously cropped with potatoes.

Diagnosis and Identification:

- Visual diagnosis based on narrow dark brown stem and petiole streaks, small leaf spots, downward leaf curling, and greasy rough fruit lesions.
- Laboratory diagnostic techniques are not specified in the provided sources.

Prevention and Cultural Practices:

- Using ToMV-resistant tomato varieties in combination with roguing (removing) infected plants.
- Avoiding handling potatoes prior to working with tomato plants.
- Avoiding planting tomatoes in fields that were planted with potatoes in the previous season.
- Sterilizing workers' tools after handling infected plants.

Biological and Chemical Control:

- Chemical pesticide names, doses, application methods, and biological control agents are not specified in the provided sources.

Integrated Disease Management and Monitoring:

- Integrating genetic host resistance (ToMV resistance), crop rotation excluding potatoes, tool sterilization, and roguing.
- Regular field inspection of stems and foliage for necrotic streaks and brown lesions.

Effects on Crop Growth, Fruit Quality, and Yield:

- Causes widespread leaf necrosis and stem streaking, impairing plant vigor.
- Induces greasy, rough, misshapen lesions on fruits, making them unmarketable and reducing total yield.

Other Relevant Information:

- Geographic distribution is worldwide.

1.41 Tomato Mottle Virus

Disease Name and Causal Organism:

- Disease Name: Tomato Mottle Virus (Tomato Mottle).
- Causal Organism: Tomato mottle virus (ToMoV), a geminivirus.

Symptoms and Signs:

- Stunting and reduced plant growth, which are more pronounced when plants are infected at an early stage.
- Upward curling and distortion of upper and middle leaves.
- Chlorosis or yellowing of foliage.
- Lack of fruit set and reduced fruit size.

Disease Development and Spread:

- Transmitted easily and efficiently by the silverleaf whitefly vector (*Bemisia argentifolii*).
- Has a fairly narrow host range including common bean, tropical soda apple, and a few solanaceous weeds (does not infect bell pepper).
- Not easily mechanically transmitted; secondary spread within tomato crops occurs via whiteflies.
- Severe disease outbreaks are associated with large whitefly populations.

Conditions Favourable for Disease Development:

- High population density of the silverleaf whitefly vector.
- Presence of nearby alternative hosts such as tropical soda apple, common bean, or solanaceous weeds.

Diagnosis and Identification:

- Visual identification based on stunted growth, upward leaf curling, interveinal foliage yellowing, and whitefly infestation.
- Diagnostic laboratory procedures are not specified in the provided sources.

Prevention and Cultural Practices:

- Controlling solanaceous weeds and alternative hosts in the vicinity of tomato fields.
- Planting tomato crops during times of the year that avoid peak whitefly populations.

Biological and Chemical Control:

- Regular use of mineral oil sprays may reduce virus spread rate by reducing acquisition and infection by whiteflies.

- Regular application of insecticides provides some vector control; however, whiteflies can develop pesticide resistance if insecticides are used excessively.
- Specific chemical names, doses, or biological control agents are not specified in the provided sources.

Integrated Disease Management and Monitoring:

- Combining weed host sanitation, seasonal planting adjustments, mineral oil sprays, and judicious insecticide use.
- Monitoring whitefly vector populations and early visual symptoms of foliage distortion.

Effects on Crop Growth, Fruit Quality, and Yield:

- Causes severe plant stunting and growth reduction.
- Significantly lowers fruit yield due to failure of fruit set and smaller fruit size.

Other Relevant Information:

- Geographic distribution is the Southeastern USA (first recognized in Florida in 1989).